



College of Engineering and Physical Sciences

Department of Physics

PHYS 498 – Senior Project II

SRP II Final Report:

Investigations of Al/Al₂O₃ Multilayer Stacks for Metamaterial Applications in the
UV Range of Frequencies

Student Name: Adel Hamed Al-yoorby

ID: 100063397

Supervisor Name: Dr. Dalaver H. Anjum (Advisor)

Dr. Davide Batic (Co-Advisor)

Dr. Denys Dutykh (Co-Advisor)

Semester: Spring 2026

Date: 24 April 2026

Abstract

This work investigates the electromagnetic and optical behavior of Al/Al₂O₃ multilayer thin-film stacks for potential metamaterial applications in the ultraviolet (UV) regime. Finite-difference time-domain (FDTD) simulations are first employed to evaluate the idealized response of a metal–dielectric multilayer, demonstrating negative refraction under oblique incidence.

Multilayer structures are then fabricated using electron-beam evaporation with controlled oxidation, forming alternating aluminum (~5 nm) and aluminum oxide (~7 nm) layers. Structural and compositional characterization using transmission electron microscopy (TEM), high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM), and energy-dispersive X-ray spectroscopy (EDS) confirms the formation of a periodic multilayer system, though with non-ideal interfaces and thickness variations.

Dielectric properties are reconstructed via low-loss STEM-electron energy-loss spectroscopy (EELS), revealing a pronounced plasmon resonance near ~15 eV and a zero-crossing in the real permittivity in the deep-UV region. However, hyperbolic behavior is not observed within the measured spectral range. Optical characterization using UV–Vis reflectance, photoluminescence (PL), and cathodoluminescence (CL) indicates strong absorption and weak radiative emission, consistent with a highly lossy system.

Overall, the fabricated Al/Al₂O₃ multilayer captures key plasmonic characteristics relevant to UV metamaterials. While simulations demonstrate that negative refraction is achievable in ideal structures, experimental performance is limited by optical losses and structural non-idealities.

1) Introduction

1.1) Background and Motivation

Metamaterials are artificially engineered structures designed to exhibit electromagnetic properties not attainable in naturally occurring materials. Unlike conventional media, whose optical response is governed primarily by intrinsic electronic structure, metamaterials derive their behavior from subwavelength structuring, enabling control over effective dielectric properties and wave propagation [1-3]. This allows access to phenomena such as negative refraction, anisotropic propagation, and sub-diffraction optical confinement [2-5].

A key manifestation of such engineered electromagnetic response is negative refraction, in which an incident electromagnetic wave refracts on the opposite side of the interface normal compared to conventional materials. This behavior arises from a reversal of phase propagation associated with negative refractive index media [1-2]. The concept is illustrated in Fig. 1, which compares light propagation in conventional materials and metamaterials, highlighting the contrasting refraction behavior.

Beyond isotropic negative-index systems, a broader and more practical class of materials is provided by hyperbolic metamaterials (HMMs). These media are characterized by anisotropic dielectric tensors whose principal components have opposite signs, enabling hyperbolic dispersion and the propagation of high- k modes that are evanescent in conventional materials [4-7]. Such behavior enables enhanced light-matter interaction and access to subwavelength spatial information, making HMMs attractive for applications in nanoscale photonics.

However, extending these capabilities into the ultraviolet (UV) regime remains a significant challenge. Conventional plasmonic materials such as gold (Au) and silver (Ag) exhibit strong interband transitions at high photon energies, leading to increased optical losses and damping of plasmonic excitations [8-9]. These losses limit field confinement and suppress metamaterial behavior at shorter wavelengths.

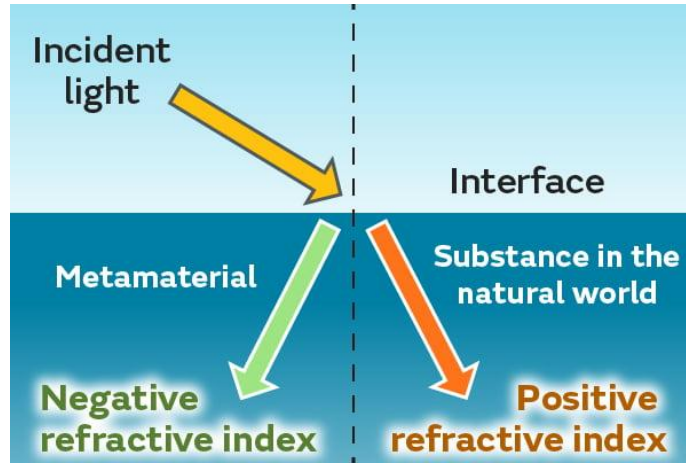


Fig. 1. Schematic illustration of light refraction at an interface between a conventional material (positive refractive index) and a metamaterial (negative refractive index). The incident beam refracts on opposite sides of the interface normal depending on the sign of the refractive index. Adapted from [13].

1.2) Motivation for Multilayer Metamaterials

To overcome these limitations, metal–dielectric multilayer structures have emerged as a practical platform for realizing metamaterial behavior. By alternating metallic and dielectric layers at subwavelength scales, these systems can be engineered to exhibit anisotropic electromagnetic response and tunable optical properties [4, 6].

Such multilayer stacks offer several advantages:

- Flexibility in tuning optical response through geometry
- Compatibility with thin-film fabrication techniques
- Ability to approximate anisotropic effective media

These characteristics make multilayer systems a promising route for achieving metamaterial functionality in challenging spectral regimes such as the UV.

1.3) Material System: Al/Al₂O₃ Multilayers

Among candidate material systems, Al/Al₂O₃ multilayers are particularly attractive for UV applications. Aluminum provides plasmonic response at higher photon energies compared to noble metals, while aluminum oxide serves as a stable dielectric spacer that preserves electromagnetic coupling between layers [10-12].

This combination enables the formation of metal–dielectric stacks capable of supporting

anisotropic optical response, while remaining compatible with established nanofabrication techniques.

1.4) Material System: Al/Al₂O₃ Multilayers

Among candidate material systems, Al/Al₂O₃ multilayers are particularly attractive for UV applications. Aluminum provides plasmonic response at higher photon energies compared to noble metals, while aluminum oxide serves as a stable dielectric spacer that preserves electromagnetic coupling between layers [10-12].

This combination enables the formation of metal–dielectric stacks capable of supporting anisotropic optical response, while remaining compatible with established nanofabrication techniques.

1.5) Scope and Relevance of This Work

The present study investigates Al/Al₂O₃ multilayer stacks as a platform for UV metamaterial behavior, focusing on the relationship between structure, dielectric response, and optical properties.

Rather than assuming idealized behavior, this work emphasizes experimental validation, addressing key challenges such as:

- structural non-uniformity
- interface roughness
- optical losses

By combining multiple characterization techniques with electromagnetic simulations, the study aims to establish a comprehensive understanding of how real multilayer systems deviate from ideal metamaterial predictions.

1.6) Objective of This Work

The primary objective of this work is to evaluate whether Al/Al₂O₃ multilayer stacks can exhibit metamaterial-relevant electromagnetic behavior in the ultraviolet regime.

Specifically, the study aims to:

- Characterize structural and compositional properties of the multilayer system

- Reconstruct dielectric response using STEM-EELS
- Analyze optical behavior using UV–Vis, PL, and CL spectroscopy
- Compare experimental results with FDTD simulations of idealized structures

Through this approach, the work seeks to determine the extent to which real multilayer systems approach ideal metamaterial behavior and to identify the key limitations governing their performance.

2) Literature Review and Background

2.1) Plasmonic Response in Metals

The interaction of electromagnetic radiation with metals is governed by the collective motion of conduction electrons. In metallic systems, these electrons can be approximated as a free-electron gas that responds coherently to incident electromagnetic fields. This collective response forms the physical basis of plasmonic phenomena [11, 19, 20].

When an electromagnetic wave interacts with a metal, the conduction electrons oscillate collectively relative to the positively charged ionic background. These oscillations give rise to charge density waves that can couple with electromagnetic fields, forming quasiparticles known as plasmons [20-21].

The optical response of a material is described by its complex dielectric function:

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \quad (1)$$

where $\varepsilon_1(\omega)$ represents the dispersive component of the response, and $\varepsilon_2(\omega)$ accounts for absorption and energy dissipation within the material [11, 19].

For plasmonic behavior to occur, the real part of the dielectric function must become negative:

$$\varepsilon_1(\omega) < 0 \quad (2)$$

This condition enables electromagnetic waves to be confined at metal interfaces and supports the excitation of surface-bound modes [11,20]. Physically, a negative real permittivity indicates that the electron response dominates over the restoring force of the lattice, allowing collective oscillations to persist [14].

Plasmonic excitations in metals manifest in two principal forms:

- Surface Plasmon Polaritons (SPPs): These are propagating electromagnetic modes bound to metal–dielectric interfaces. They arise from the coupling between electromagnetic waves and collective electron oscillations, and they exhibit dispersion relations that depend on both the metal

and dielectric properties [20,22].

- **Localized Surface Plasmons (LSPs):** These occur in nanoscale metallic structures, where the electron oscillations are spatially confined. LSPs do not propagate but instead produce strong localized electric fields and resonant scattering behavior that depends on particle size, shape, and surrounding environment [20,22].

The distinction between propagating and localized plasmonic modes is illustrated in Fig. 2, where SPPs exhibit wave-like propagation along an interface, while LSPs show discrete resonant behavior associated with confined geometries.

In addition to the dielectric function, plasmonic resonances can be characterized using the energy-loss function, defined as:

$$\text{Im}\left(-\frac{1}{\varepsilon(\omega)}\right) \quad (3)$$

This quantity represents the probability that a fast electron loses energy through excitation of collective modes within the material. Peaks in the energy-loss function correspond to plasmon resonances, providing a direct connection between electronic structure and measurable optical response [21-22].

This formalism is particularly important in experimental techniques such as electron energy-loss spectroscopy (EELS), where plasmonic excitations are directly observed and used to reconstruct the dielectric function of materials at the nanoscale.

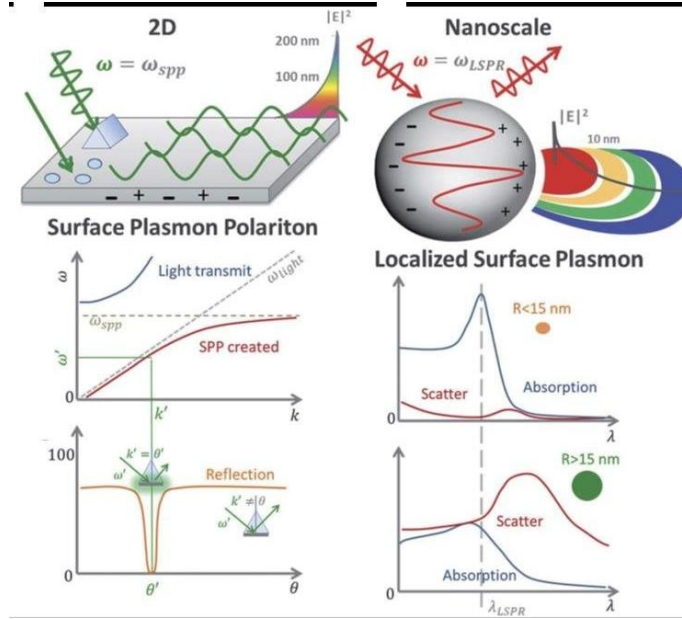


Fig. 2. Illustration of plasmonic modes in metallic systems. Surface plasmon polaritons (SPPs) propagate along metal–dielectric interfaces with characteristic dispersion behavior, while localized surface plasmons (LSPs) arise from confined electron oscillations in nanoscale structures, exhibiting size-dependent scattering and absorption responses. Adapted from [27].

2.2) Aluminum Plasmonics

Building on the plasmonic framework introduced in Section 2.1, aluminum represents a distinct class of plasmonic material due to its ability to sustain plasmonic behavior at significantly higher photon energies. This is primarily a consequence of its high plasma frequency, which lies near ~ 15 eV [10,23]. As a result, the spectral range over which aluminum exhibits negative real permittivity extends into the ultraviolet (UV), unlike noble metals such as gold and silver, whose plasmonic response is strongly limited by interband transitions at high photon energies [8,10]. This characteristic makes aluminum particularly suitable for applications requiring controlled electromagnetic response at short wavelengths, including ultraviolet photonics and nanoscale optical systems [10,12].

The tunability of aluminum plasmonic response is illustrated in Fig. 3, which presents both experimental and theoretical scattering spectra for nanostructures of varying size. As the characteristic dimension of the aluminum structures increases, the plasmon resonance shifts toward longer wavelengths, demonstrating that the optical response is strongly dependent on nanoscale geometry [10]. This behavior highlights the ability to engineer the spectral position and intensity of plasmonic resonances through structural design, which is essential for tailoring electromagnetic response in practical systems. The observed agreement between experimental measurements and theoretical predictions further confirms that aluminum-based plasmonic systems can be reliably modeled and optimized for specific optical applications.

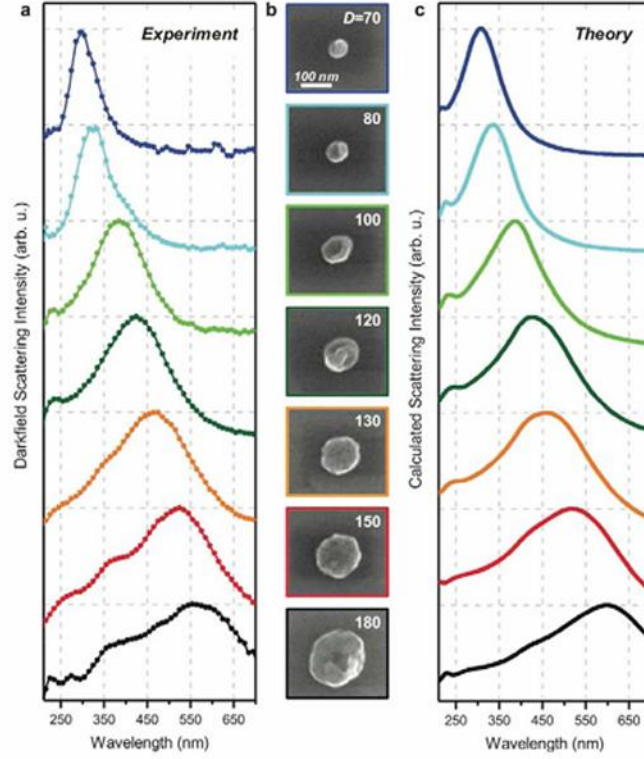


Fig. 3. Experimental and theoretical optical response of aluminum nanostructures of varying size. Measured and simulated scattering spectra show size-dependent plasmon resonance shifts, demonstrating tunable plasmonic behavior in aluminum. Adapted from [10].

The dependence of plasmonic behavior on geometry arises from the interaction between conduction electrons and the surrounding electromagnetic field, which is influenced by both the dielectric environment and structural confinement [20,22]. Larger structures support longer-wavelength resonances due to increased effective oscillation length, while smaller structures exhibit stronger confinement and higher-energy resonances. This size-dependent behavior is particularly relevant for engineered systems such as multilayer stacks, where nanoscale dimensions play a dominant role in determining optical response.

Despite its advantages, aluminum exhibits significant optical losses due to intrinsic damping mechanisms. These include electron–phonon scattering, surface scattering in nanoscale structures, grain boundaries, and the presence of oxide layers at interfaces [12,23]. These effects increase the imaginary component of the dielectric function, leading to broader plasmon resonances and reduced electromagnetic field enhancement. As a result, while aluminum enables plasmonic behavior in the UV regime, its performance is ultimately limited by absorption losses that must be considered in practical applications.

In the context of metal–dielectric multilayer systems, aluminum serves as the plasmonic component, providing the negative permittivity required for engineered electromagnetic response.

When combined with a dielectric such as Al_2O_3 , the contrast between metallic and dielectric layers enables the formation of anisotropic effective media capable of supporting metamaterial-like behavior [4,6]. This metal–dielectric contrast is fundamental to the multilayer structures investigated in this work and forms the basis for the dielectric and electromagnetic properties analyzed in subsequent sections.

2.3) Role of Aluminum Oxide (Al_2O_3)

In metal–dielectric multilayer systems, aluminum oxide (Al_2O_3) serves as the dielectric counterpart to the plasmonic aluminum layers, providing both structural and optical functionality. As a wide-bandgap material with a bandgap typically in the range of 7–9 eV, Al_2O_3 exhibits low absorption across the ultraviolet–visible spectral range and maintains a positive real permittivity ($\epsilon > 0$) under normal operating conditions [9,24]. This makes it an effective dielectric spacer that introduces minimal additional loss while preserving electromagnetic coupling within the multilayer structure.

From an electromagnetic perspective, the role of Al_2O_3 is primarily defined by its contrast with the metallic aluminum layers. While aluminum provides a negative real permittivity necessary for plasmonic response, Al_2O_3 contributes a positive permittivity, creating the conditions required for anisotropic effective media in layered systems [4,6]. This contrast is fundamental to the formation of engineered dielectric responses, particularly in multilayer configurations where alternating materials determine the overall optical behavior.

In addition to its dielectric properties, Al_2O_3 plays a critical role in defining the structural integrity of the multilayer stack. During fabrication, the oxide layer forms through controlled oxidation of deposited aluminum, creating thin dielectric spacers between metallic layers. These layers not only separate adjacent metal films but also define the interfaces that govern electromagnetic coupling across the structure. However, in practical systems, the oxidation process can lead to interface broadening and compositional intermixing, which may deviate from the idealized sharp interfaces assumed in theoretical models [24,25].

Furthermore, Al_2O_3 contributes to the optical response through the presence of defect and interface states. In particular, oxygen vacancies and other point defects within the oxide layer can introduce localized energy states within the bandgap, acting as centers for radiative recombination [24,25]. These defect-related states are often associated with emission in the visible range and are relevant to the photoluminescence (PL) and cathodoluminescence (CL) behavior observed in metal–oxide systems. As a result, Al_2O_3 does not function purely as a passive dielectric, but also plays an active role in the optical characteristics of the multilayer structure.

In the context of the present work, the inclusion of Al_2O_3 is therefore essential for both structural and electromagnetic reasons. Structurally, it defines the multilayer architecture and interfaces, while electromagnetically, it provides the positive-permittivity component required to establish metal–dielectric contrast. This dual role makes Al_2O_3 a key element in enabling anisotropic optical response and in determining the overall performance of Al/ Al_2O_3 multilayer systems.

2.4) Internal Structure

To Metal–dielectric multilayer structures consist of alternating thin films of metallic and dielectric

materials arranged with nanoscale periodicity. When the characteristic thickness of each layer is significantly smaller than the wavelength of the incident electromagnetic radiation, the structure can be approximated as a homogenized medium with anisotropic dielectric properties [4,6,15]. This approximation is valid in the long-wavelength limit, where the electromagnetic field does not resolve individual layers but instead interacts with an effective medium defined by the overall structure.

A fundamental parameter governing the electromagnetic response of such multilayer systems is the **filling fraction**, defined as the ratio of the metal layer thickness to the total bilayer thickness:

$$f = \frac{d_m}{d_m + d_d} \quad (4)$$

where d_m and d_d denote the thicknesses of the metal and dielectric layers, respectively [4,6]. The filling fraction determines the relative contribution of each material to the effective dielectric response and therefore plays a central role in controlling the optical behavior of the multilayer.

In addition to the filling fraction, the number of periods N defines the total thickness of the multilayer stack:

$$L = N(d_m + d_d) \quad (5)$$

where L is the total thickness of the structure. While the filling fraction directly influences the effective dielectric properties, the number of periods governs how well the finite multilayer approximates a bulk anisotropic medium. Increasing N reduces the influence of surface effects and improves the validity of the effective medium approximation [6,15,18].

The electromagnetic response of multilayer systems arises from the contrast between the dielectric properties of the constituent materials. In metal–dielectric stacks, the combination of a negative-permittivity metal and a positive-permittivity dielectric produces an anisotropic response that cannot be achieved in homogeneous media. This anisotropy originates from the layered geometry, which causes the electromagnetic field to interact differently depending on its orientation relative to the layers.

For fields oriented parallel to the layers, the electric field experiences the materials as effectively connected in parallel, whereas for fields perpendicular to the layers, the response is governed by continuity conditions across the interfaces. These differences lead to direction-dependent dielectric behavior, forming the basis for anisotropic optical properties [6,15,16].

As illustrated in Fig. 4, the anisotropic nature of metal–dielectric multilayers results in modified wave-vector propagation and non-circular isofrequency contours. Under appropriate conditions, this can lead to unconventional electromagnetic behavior, including negative refraction. In such cases, the transmitted wave bends to the opposite side of the interface normal compared to conventional materials, demonstrating the impact of engineered anisotropy on wave propagation

[26].

However, the behavior predicted by idealized models is not always fully realized in experimental systems. Practical multilayer structures are subject to several limitations, including interface roughness, thickness variation, and compositional intermixing. These non-idealities modify the effective dielectric response and introduce additional optical losses, reducing the degree of anisotropy and limiting the propagation of high- k modes [17-18].

Furthermore, the finite number of layers in experimental samples introduces deviations from the bulk effective medium approximation. When the number of periods is limited, surface and finite-size effects become significant, altering the electromagnetic response and potentially suppressing ideal metamaterial behavior. As a result, while multilayer systems can approach the properties predicted by effective medium theory, their performance is ultimately constrained by fabrication and material limitations.

In the present work, the $\text{Al}/\text{Al}_2\text{O}_3$ multilayer system is designed with nanometer-scale layer thicknesses, ensuring operation in the deep subwavelength regime. Under these conditions, the structure approximates an anisotropic effective medium while still retaining the influence of non-ideal interfaces and finite-size effects. This balance between idealized behavior and experimental reality is critical for interpreting the results presented in later sections.

2.5) Hyperbolic Metamaterials

Hyperbolic metamaterials represent a class of anisotropic media in which the principal components of the dielectric permittivity tensor possess opposite signs. In layered metal–dielectric systems, this behavior arises from the contrast between the negative permittivity of the metallic component and the positive permittivity of the dielectric component, leading to a direction-dependent electromagnetic response [4-5]. The condition for hyperbolic behavior is expressed as

$$\text{Re}(\varepsilon_{\parallel}) \cdot \text{Re}(\varepsilon_{\perp}) < 0 \quad (6)$$

where $\varepsilon_{\parallel}$ and $\varepsilon_{\perp}$ are the effective permittivity components parallel and perpendicular to the multilayer interfaces, respectively [4-6]. When this condition is satisfied, the isofrequency surface in momentum space transitions from a closed (elliptic) geometry to an open (hyperbolic) geometry, fundamentally altering the propagation characteristics of electromagnetic waves.

This change in dispersion can be visualized in Fig. 4, where the anisotropic structure of the multilayer leads to non-circular isofrequency contours. In conventional materials, wave propagation is governed by isotropic dispersion relations, resulting in symmetric wave-vector distributions. In contrast, hyperbolic media support highly directional propagation due to the open nature of the dispersion surface, enabling access to high- k modes that are evanescent in ordinary

materials [4-5].

The physical consequence of this behavior is the ability of hyperbolic metamaterials to support electromagnetic waves with very large wave vectors, corresponding to subwavelength spatial features. This leads to several unique optical properties, including enhanced photonic density of states, increased spontaneous emission rates, and the possibility of imaging beyond the diffraction limit [4-5,7].

In addition to supporting high- k modes, anisotropic multilayer systems can exhibit negative refraction, where the transmitted wave vector bends to the opposite side of the interface normal compared to conventional materials. As illustrated in Fig. 4, the direction of energy propagation is determined by the shape of the dispersion relation rather than by a simple scalar refractive index. This results in unconventional wave propagation behavior that is fundamentally linked to the anisotropic dielectric response of the medium [26].

However, the realization of hyperbolic behavior in practical systems is strongly influenced by material losses and structural imperfections. The presence of a significant imaginary component of the permittivity (ϵ_2) leads to damping of high- k modes, reducing their propagation length and limiting their experimental observability [17-18]. Additionally, deviations from ideal periodicity, such as interface roughness and compositional broadening, can alter the effective dielectric tensor and weaken the conditions required for hyperbolic dispersion.

As a result, while multilayer systems may theoretically satisfy the hyperbolic condition in Eq. (6), experimental observation of clear hyperbolic behavior is often limited to specific spectral regions. In many cases, systems exhibit plasmonic characteristics without fully entering a well-defined hyperbolic regime, particularly when losses dominate the optical response.

In the context of the present work, the $\text{Al}/\text{Al}_2\text{O}_3$ multilayer system is expected to approach hyperbolic behavior due to the strong contrast between its constituent materials. However, as will be shown in subsequent sections, experimental measurements indicate that while plasmonic behavior is clearly observed, the conditions required for fully developed hyperbolic dispersion are not satisfied within the measured spectral range. This highlights the importance of both material properties and structural quality in determining the electromagnetic response of multilayer metamaterials.

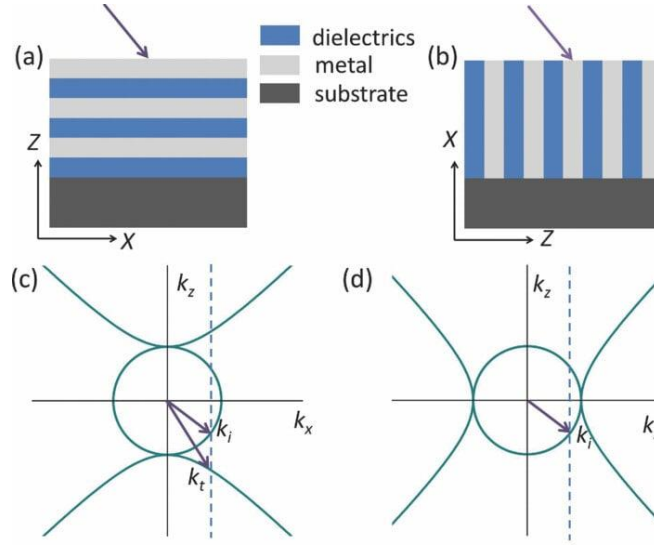


Fig. 4. Schematic representation of metal–dielectric multilayer structures and their corresponding dispersion behavior. Panels (a) and (b) illustrate layered geometries with different orientations relative to the propagation direction. Panels (c) and (d) show the corresponding isofrequency contours in momentum space, highlighting the transition from elliptic to hyperbolic dispersion and the resulting modification of wave-vector propagation. Adapted from [26].

2.6) Effective Medium Approximation

The electromagnetic behavior of metal–dielectric multilayer systems can be quantitatively described using the effective medium approximation (EMA), which treats the layered structure as a homogenized anisotropic material when the layer thickness is much smaller than the wavelength of the incident radiation [6,15,16]. Under this approximation, the discrete multilayer is replaced by an equivalent medium characterized by an effective dielectric tensor, whose components depend on the properties and relative proportions of the constituent materials.

For a lamellar metal–dielectric stack, the effective permittivity components parallel and perpendicular to the layers are given by [6,15,16]

$$\varepsilon_{\parallel} = f\varepsilon_m + (1 - f)\varepsilon_d \quad (7)$$

$$\varepsilon_{\perp} = \left(\frac{f}{\varepsilon_m} + \frac{1-f}{\varepsilon_d} \right)^{-1} \quad (8)$$

where ε_m and ε_d are the permittivities of the metal and dielectric, respectively, and f is the filling fraction defined previously in Eq. (4). These expressions arise from solving Maxwell's equations under the boundary conditions imposed by the layered geometry. Specifically, Eq. (7) corresponds to the case where the electric field is oriented parallel to the layers, resulting in an arithmetic averaging of the permittivities, while Eq. (8) corresponds to the perpendicular direction, where continuity of the displacement field leads to a harmonic averaging [6,15,16].

These relations demonstrate that the effective dielectric response is highly sensitive to both the intrinsic material properties and the geometrical configuration of the multilayer. In particular, when the metal exhibits negative permittivity and the dielectric remains positive, the effective tensor components can acquire opposite signs, satisfying the hyperbolic condition defined in Eq. (6). This provides the theoretical basis for achieving hyperbolic metamaterial behavior in metal–dielectric multilayer systems.

However, the validity of the effective medium approximation is subject to several important limitations. First, the approximation assumes that the electromagnetic field varies slowly over the scale of the multilayer period, which requires that the layer thicknesses be much smaller than the wavelength. In the present system, where layer thicknesses are on the order of a few nanometers, this condition is generally satisfied in the UV–visible regime. Nevertheless, deviations can arise when the wavelength approaches the scale of the multilayer period, reducing the accuracy of the homogenization [6,15].

Second, EMA assumes perfectly periodic structures with sharp interfaces and uniform layer thickness. In experimentally fabricated multilayers, deviations from these ideal conditions—such as interface roughness, interdiffusion, and compositional broadening—modify the local dielectric environment and lead to deviations from the predicted effective response [17–18]. These effects are particularly important in nanoscale systems, where even small structural variations can significantly influence optical behavior.

Third, the presence of material losses, represented by the imaginary component of the permittivity, plays a critical role in determining the observable electromagnetic response. In lossy systems, the propagation of high- k modes predicted by EMA is strongly attenuated, limiting their contribution to measurable optical phenomena [17]. As a result, while the effective medium model may predict hyperbolic behavior, the experimental manifestation of such behavior can be suppressed by absorption.

Finally, the finite number of layers in real multilayer systems introduces additional deviations from the idealized infinite medium assumed in EMA. When the number of periods is limited, boundary effects and surface modes become significant, altering the overall electromagnetic response and potentially masking bulk anisotropic behavior [18].

In the context of the present work, the effective medium approximation provides a useful theoretical framework for interpreting the dielectric response reconstructed from STEM-EELS measurements and for understanding the electromagnetic behavior observed in FDTD simulations. However, as will be discussed in later sections, the experimentally measured system deviates from the ideal EMA predictions due to structural non-uniformity and optical losses. This highlights the importance of combining theoretical models with experimental characterization to obtain an accurate description of real multilayer metamaterials.

3) Methodology

The methodology employed in this study is designed to investigate the electromagnetic and optical behavior of Al/Al₂O₃ multilayer stacks through a combined numerical and experimental approach. The analysis follows a hierarchical framework in which electromagnetic modeling is first used to establish the expected behavior of an idealized system, followed by fabrication and multi-scale characterization to evaluate the properties of the experimentally realized structure. This approach enables a direct correlation between structural, compositional, dielectric, and optical properties.

3.1 Electromagnetic Modeling (FDTD)

The electromagnetic response of the multilayer system is modeled using the finite-difference time-domain (FDTD) method, a numerical technique for solving Maxwell's equations in space and time for complex geometries [28-31]. In this approach, the time-dependent Maxwell curl equations are discretized:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}, \quad \nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad (9)$$

where $\mathbf{E}$ and $\mathbf{H}$ represent the electric and magnetic fields, respectively, and ε and μ denote the material permittivity and permeability [28,29]. The equations are solved using the Yee grid, which staggers field components in space and time to ensure numerical stability and second-order accuracy [29].

The simulated structure consists of a periodic stack of alternating aluminum and aluminum oxide layers. An idealized symmetric configuration is used, with equal layer thicknesses of approximately 7 nm for both materials, enabling direct comparison with effective medium theory predictions [6,15].

A monochromatic Gaussian beam is introduced at an oblique angle of incidence (typically 45°), allowing investigation of anisotropic wave propagation and potential negative refraction behavior [32,33]. Perfectly matched layer (PML) boundary conditions are applied to eliminate artificial reflections at the simulation boundaries [30,31].

Material properties are incorporated using frequency-dependent dielectric functions obtained from literature sources [8-10]. Aluminum is modeled as a dispersive, lossy medium, while Al₂O₃ treated as a low-loss dielectric [9,10]. The primary output of the simulation is the spatial distribution of the electric field intensity, which provides insight into wave propagation and electromagnetic response within the multilayer.

3.2 Nanofabrication of Multilayer Stacks

The Al/Al₂O₃ multilayer structures are fabricated using electron-beam evaporation, a physical vapor deposition technique widely used for the growth of thin films with nanometer-scale thickness control [34,35]. In this process, a focused electron beam is used to heat and evaporate a

source material in a high-vacuum environment, allowing the vaporized atoms to condense onto a substrate.

The multilayer architecture is formed through alternating deposition of aluminum layers and controlled oxidation steps. Following deposition, aluminum is exposed to oxygen, forming aluminum oxide (Al_2O_3) due to its high chemical reactivity [36]. This results in a sequence of metal–dielectric layers without the need for independent oxide deposition.

The fabricated structure consists of:

- aluminum layers of approximately 5 nm thickness
- Al_2O_3 spacer layers of approximately 7 nm thickness

resulting in a bilayer periodicity:

$$d = d_m + d_d \approx 12 \text{ nm} \quad (10)$$

This configuration satisfies the subwavelength condition $d \ll \lambda$, allowing the multilayer to approximate an effective medium as described in Section 2 [6,15].

3.3 Structural Characterization

The structural properties of the multilayer system are investigated using transmission electron microscopy (TEM), which provides cross-sectional imaging of thin-film structures at nanometer resolution [37]. TEM enables direct observation of layer continuity, thickness, and overall morphology.

Cross-sectional samples are prepared using focused ion beam (FIB) milling, a technique that allows precise extraction of thin lamellae for nanoscale imaging [38]. This preparation method ensures that the internal structure of the multilayer can be examined without significant distortion.

TEM analysis provides the initial verification of the multilayer architecture, serving as the foundation for subsequent high-resolution and compositional characterization.

3.4 HAADF-STEM Characterization

High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) is used to obtain high-resolution structural information with enhanced compositional contrast. In HAADF imaging, the detected signal intensity scales approximately with the atomic number Z , producing so-called Z -contrast images [39].

As a result:

- aluminum-rich regions appear brighter

- oxide regions appear darker

This contrast allows direct visualization of the alternating metal–dielectric layers, as well as identification of interface morphology and thickness variation at the nanoscale. HAADF-STEM therefore complements TEM by providing detailed structural information beyond conventional imaging.

3.5 Compositional Analysis (EDS)

Energy-dispersive X-ray spectroscopy (EDS) is used to determine the elemental composition of the multilayer structure. When the sample is irradiated with an electron beam, characteristic X-rays are emitted from each element, allowing identification based on energy signatures [40].

EDS mapping provides spatial distribution of aluminum and oxygen across the multilayer, confirming the periodic structure and revealing compositional variations at interfaces. Line profiles extracted from EDS data enable quantitative analysis of elemental gradients, providing insight into interface sharpness and compositional broadening.

3.6 Dielectric Reconstruction (STEM-EELS)

The dielectric response of the multilayer system is reconstructed using scanning transmission electron microscopy–electron energy-loss spectroscopy (STEM-EELS). In this technique, fast electrons interact with the sample and lose energy through excitation of collective electronic modes [21-22].

The measured energy-loss spectrum is related to the dielectric function through:

$$\text{Im}\left(-\frac{1}{\varepsilon(\omega)}\right) \quad (3)$$

Using Kramers–Kronig analysis, the real and imaginary components of the dielectric function can be extracted from the measured spectra [21]. This allows direct determination of:

- plasmon resonance energy
- dielectric dispersion
- electronic response of the multilayer

This technique provides a critical link between nanoscale structure and macroscopic electromagnetic behavior.

3.7 Optical Characterization

The optical response of the multilayer system is investigated using photoluminescence (PL), cathodoluminescence (CL), and UV–visible reflectance spectroscopy.

Photoluminescence measures light emitted following optical excitation, where electrons are promoted to higher energy states and subsequently recombine radiatively, producing characteristic emission spectra [41]. Cathodoluminescence uses high-energy electron excitation to generate electron–hole pairs, providing spatially resolved emission that is sensitive to defect states and interfaces [42].

UV–visible spectroscopy measures wavelength-dependent reflectance, allowing determination of optical response across the UV–visible range. These measurements provide information about absorption behavior and spectral features associated with plasmonic excitations.

Together, these techniques probe electronic transitions, radiative recombination processes, and the overall optical behavior of the multilayer system, complementing structural and dielectric characterization.

4) Results and Discussion

The results obtained in this study are discussed in the same sequence as the methodology in order to establish a direct connection between electromagnetic modeling, structural characterization, compositional analysis, dielectric reconstruction, and optical response. The central objective of this section is to determine whether the fabricated $\text{Al}/\text{Al}_2\text{O}_3$ multilayer system exhibits metamaterial-relevant behavior in the ultraviolet (UV) spectral range, which is the primary focus of this work. While several measured features extend into the visible and higher-energy regimes, the interpretation presented here emphasizes those results that are most relevant to the UV response, particularly the evolution of the dielectric function, the presence or absence of negative permittivity in the UV, and the extent to which the experimentally realized structure approaches the idealized electromagnetic behavior predicted by simulation. Through this comparison, the discussion identifies not only the signatures of plasmonic response present in the system, but also the structural and dielectric limitations that prevent the full realization of hyperbolic metamaterial behavior within the UV wavelength range of interest.

4.1) Electromagnetic Response from FDTD Simulation

The electromagnetic response of the $\text{Al}/\text{Al}_2\text{O}_3$ multilayer system was first evaluated using finite-difference time-domain (FDTD) simulations in order to establish the expected behavior of an idealized structure. The simulated electric field distribution is shown in Fig. 5, corresponding to a frequency of 1988.42 THz (wavelength ~ 150.77 nm), which lies in the deep-ultraviolet region.

The simulation considers a Gaussian beam incident at an angle of 45° interacting with a seven-

period multilayer stack composed of alternating Al and Al_2O_3 layers, each with a thickness of 7 nm. The two vertical boundaries in Fig. 5 mark the spatial extent of the multilayer region, while the dashed horizontal line represents the interface normal. The incident and transmitted beam trajectories are indicated by solid lines.

As the incident beam enters the multilayer, a clear deviation from conventional refraction is observed. The transmitted beam bends toward the opposite side of the interface normal compared to the direction expected for positive-index materials, demonstrating negative refraction behavior. This observation indicates that, under idealized conditions, the multilayer system can exhibit metamaterial-like electromagnetic response in the deep-UV regime.

The origin of this behavior is directly linked to the optical properties of the constituent materials, shown in Fig. 6. In the ultraviolet spectral range, aluminum exhibits a strongly dispersive response, with a decreasing real component of the refractive index and a large imaginary component, indicating significant absorption. In contrast, Al_2O_3 maintains a relatively stable real refractive index with negligible absorption. This strong contrast between a lossy metallic response and a low-loss dielectric enables the formation of an effective anisotropic medium.

However, examination of Fig. 6 also reveals a key limitation: although aluminum exhibits strong dispersion, the material response in the UV is accompanied by significant losses, as indicated by the large imaginary component. These losses play a critical role in damping electromagnetic modes and limiting the propagation of high- k waves required for hyperbolic behavior.

Furthermore, the simulation assumes a perfectly periodic multilayer with uniform thickness and sharp interfaces, as well as bulk optical constants for both materials. These assumptions neglect structural imperfections and nanoscale effects present in the fabricated system. As a result, the simulated response represents an idealized upper bound, demonstrating that negative refraction is achievable in principle.

In practice, the realization of this behavior depends critically on the dielectric response within the UV spectral range. As will be shown in subsequent sections, the experimentally measured dielectric function does not fully satisfy the conditions required for hyperbolic dispersion in this region, primarily due to losses and structural non-idealities. This explains why the experimentally observed behavior deviates from the idealized simulation despite the favorable theoretical conditions.

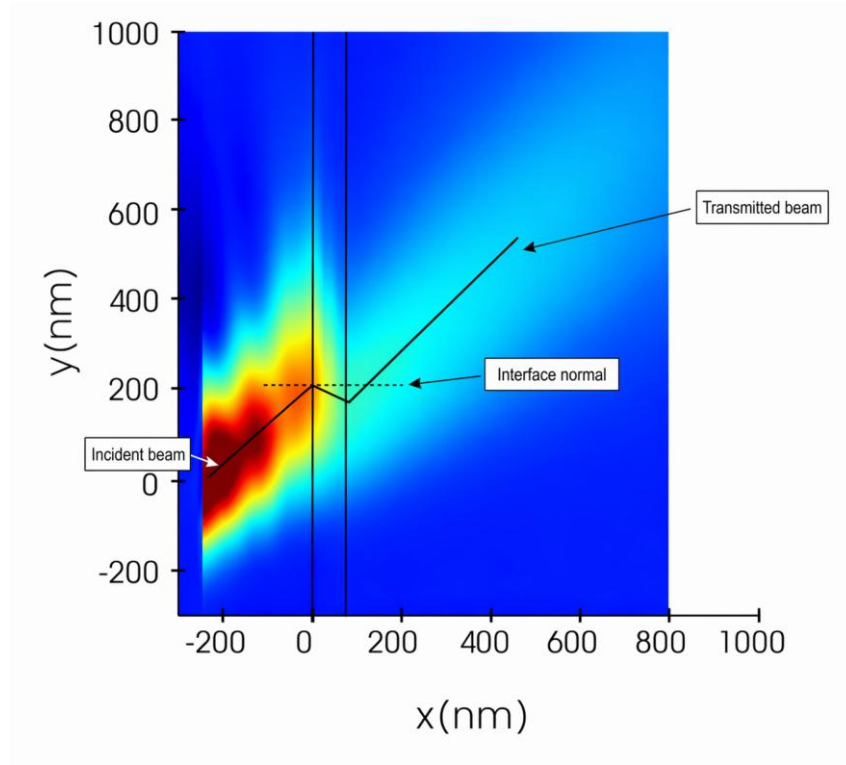


Fig. 5. Simulated P_x electric field profile at 1988.42 THz (~ 150.77 nm) for a 45° incident Gaussian beam interacting with a seven-period $\text{Al}/\text{Al}_2\text{O}_3$ multilayer (7 nm per layer). The vertical black lines indicate the boundaries of the multilayer region, and the dashed line represents the interface normal. The transmitted beam bends opposite to the expected direction for positive refraction, demonstrating negative refraction in the idealized multilayer structure.

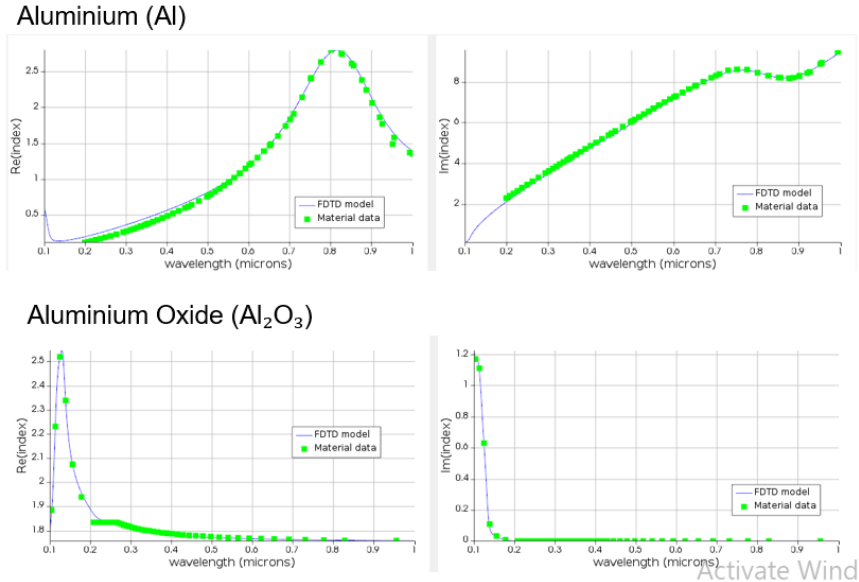


Fig. 6. Optical constants used in the FDTD simulation for aluminum (Al) and aluminum oxide (Al_2O_3). The strong contrast between the dispersive, lossy metallic response of Al and the low-loss dielectric behavior of Al_2O_3 governs the electromagnetic response of the multilayer, particularly in the ultraviolet spectral range.

4.2) Structural Results (TEM and HAADF-STEM)

The structural properties of the Al/Al_2O_3 multilayer system were investigated using transmission electron microscopy (TEM) and high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM), with representative results shown in Fig. 7. These techniques provide direct insight into the morphology, layer continuity, and interface characteristics of the multilayer structure, which are essential for understanding its electromagnetic behavior.

The cross-sectional TEM image in Fig. 7(a) confirms the formation of a continuous multilayer thin-film structure across the substrate. The layered architecture is clearly visible, indicating successful fabrication of alternating metal–dielectric regions. The overall structure maintains coherence over large lateral distances, demonstrating that the deposition process produces a stable and well-defined multilayer system.

At higher magnification, as shown in Fig. 7(b), the interfaces between layers exhibit noticeable waviness and local thickness variations. Rather than perfectly planar boundaries, the interfaces display nanoscale undulations and slight fluctuations in layer thickness. These features are characteristic of thin-film growth processes at nanometer scales and reflect the natural evolution of the film during deposition and oxidation.

In addition to interface morphology, the diffraction pattern shown in Fig. 7(c) provides information about the internal structure of the layers. The presence of distinct diffraction spots indicates the existence of crystalline aluminum-rich domains within the multilayer. This suggests that the metallic layers are not fully amorphous, but instead contain regions of crystallinity that can

influence electron scattering and optical response.

These structural characteristics are important in the context of the electromagnetic behavior discussed in Section 4.1. The FDTD simulations assume a perfectly periodic multilayer with uniform thickness and sharp interfaces. In contrast, the experimentally observed structure exhibits realistic nanoscale morphology, including interface waviness and thickness variation.

Such features influence the effective dielectric response by modifying the local metal–dielectric contrast and introducing spatial variation in the multilayer periodicity. As a result, the electromagnetic response of the system reflects both the designed multilayer architecture and the inherent structural characteristics of the fabricated films. These effects are particularly relevant in the ultraviolet regime, where the interaction between electromagnetic fields and material structure occurs on comparable length scales.

Overall, the TEM and HAADF-STEM results confirm the successful fabrication of a multilayer system while providing detailed insight into the nanoscale structural features that govern its optical and electromagnetic behavior.

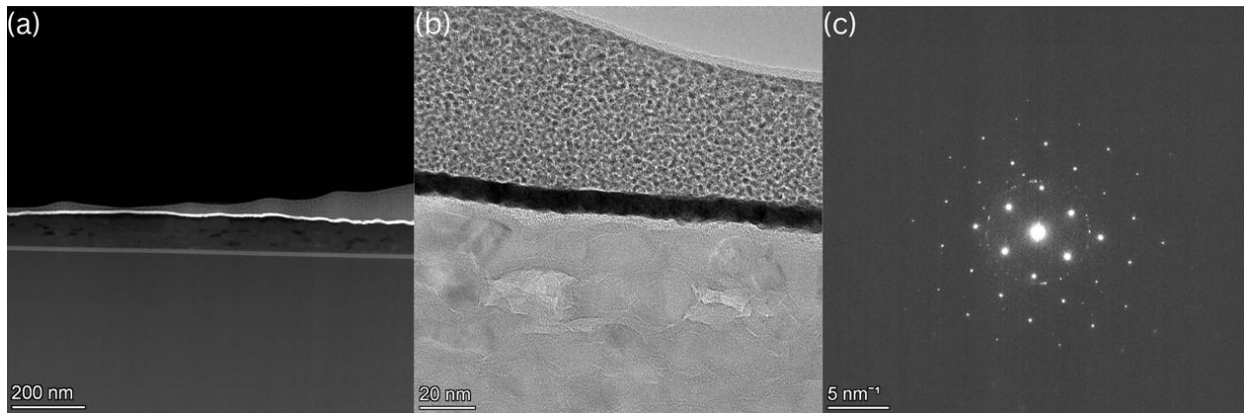


Fig. 7. Structural characterization of the Al/Al₂O₃ multilayer system. (a) Cross-sectional TEM image showing continuous multilayer thin-film structure across the substrate. (b) High-resolution image highlighting interface waviness and local thickness variations. (c) Selected area diffraction pattern indicating the presence of crystalline aluminum-rich domains.

4.3) Compositional Results (EDS)

The compositional structure of the Al/Al₂O₃ multilayer system was investigated using energy-dispersive X-ray spectroscopy (EDS), with representative elemental maps and line profiles shown in Fig. 8. These measurements provide direct information on the spatial distribution of elements across the multilayer and are essential for understanding how the material composition influences the dielectric response discussed in subsequent sections.

The elemental maps in Fig. 8(a–c) clearly distinguish between aluminum-rich and oxygen-rich regions, confirming the presence of alternating metal–dielectric layers. The periodic variation in elemental contrast across the cross-section is consistent with the intended multilayer architecture

and supports the structural observations obtained from TEM and HAADF-STEM. The aluminum signal exhibits well-defined layered regions, while the oxygen distribution corresponds to the dielectric Al_2O_3 spacers.

However, the interfaces between these regions are not sharply defined. Instead, the transition between aluminum-rich and oxygen-rich layers appears gradual, indicating compositional broadening across the interfaces. This behavior is further illustrated in the line profile shown in Fig. 8(d), where the elemental concentrations vary continuously rather than exhibiting abrupt step-like changes. Such broadening is indicative of interdiffusion and partial mixing at the metal–dielectric boundaries during fabrication.

The presence of non-abrupt interfaces has important implications for the electromagnetic response of the multilayer. In an ideal system, the dielectric function changes sharply between metal and dielectric layers, maximizing contrast and enabling strong anisotropy. In the present case, the gradual transition in composition effectively reduces this contrast, resulting in a smoother variation of the local dielectric environment.

This reduction in contrast directly influences the effective permittivity tensor described in Section 2.6. Instead of a well-defined combination of ϵ_m and ϵ_d , the system behaves as a partially graded medium, which weakens the conditions required for hyperbolic dispersion. These effects are particularly relevant in the ultraviolet spectral range, where the electromagnetic response is highly sensitive to local material composition.

Despite this, the overall periodic modulation observed in the EDS data confirms that the multilayer structure is preserved at the nanoscale. The compositional analysis therefore supports the presence of a functional metal–dielectric system, while also highlighting the realistic interface characteristics that influence its dielectric behavior.

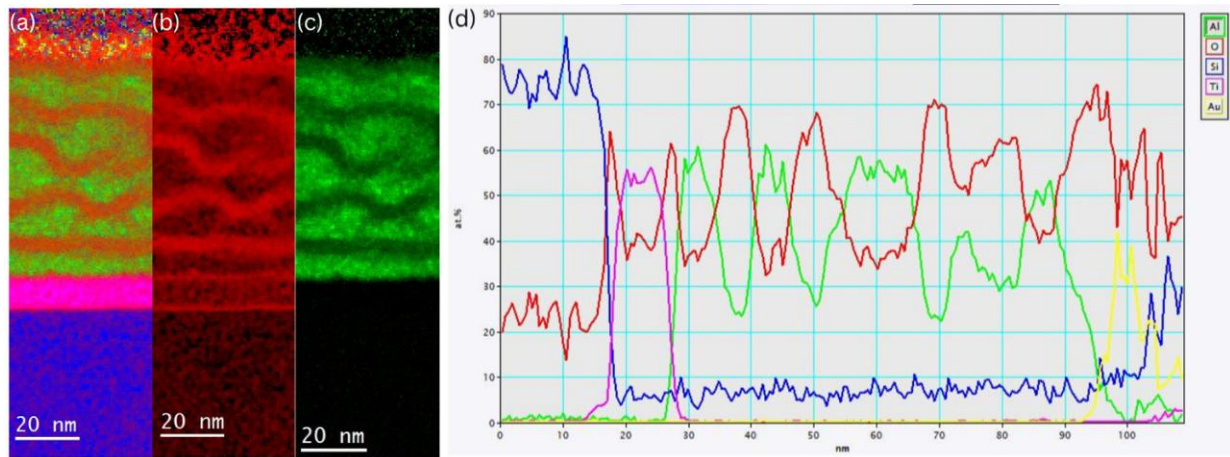


Fig. 8. Compositional characterization of the Al/ Al_2O_3 multilayer system using EDS. (a–c) Elemental maps showing spatial distribution of aluminum and oxygen across the multilayer. (d) Line profile indicating periodic compositional modulation and gradual transitions at the interfaces, consistent with interfacial broadening.

4.4) Dielectric Results (STEM-EELS)

The dielectric response of the Al/Al₂O₃ multilayer system was reconstructed using STEM-EELS measurements, with the resulting dielectric function and energy-loss spectra shown in Fig. 9. This analysis provides direct insight into the frequency-dependent electronic response of the system and is essential for evaluating its behavior within the ultraviolet spectral range (200–400 nm).

The interpretation of these results is therefore centered on this UV region. In the top panel of Fig. 9, the real part of the permittivity, ϵ_1 , remains positive throughout the near-UV range, with no indication of a zero-crossing between 200 and 400 nm. This is a key observation, as a transition to negative ϵ_1 is typically associated with plasmonic behavior. The absence of such a transition in the UV region indicates that the dielectric response remains positive within the wavelength range of interest.

The zero-crossing of ϵ_1 is observed at higher energies, near the deep-UV region, and is associated with the plasmon resonance. This is further supported by the energy-loss spectrum shown in the bottom panel of Fig. 9, which exhibits a pronounced peak near ~ 15 eV. While this peak confirms the presence of strong plasmonic activity, it occurs outside the primary UV range relevant to this study and therefore does not directly influence the electromagnetic behavior in the 200–400 nm window.

The imaginary component of the dielectric function, ϵ_2 , shown in the middle panel of Fig. 9, remains non-negligible across the UV range, indicating significant absorption. These losses contribute to damping of electromagnetic modes and play an important role in shaping the optical response of the multilayer.

Taken together, the results show that the multilayer system exhibits a well-defined dielectric response in the ultraviolet region, characterized by positive ϵ_1 and finite absorption. This behavior provides a consistent interpretation of the optical response and establishes a direct link between the measured dielectric properties and the electromagnetic behavior observed in the system.

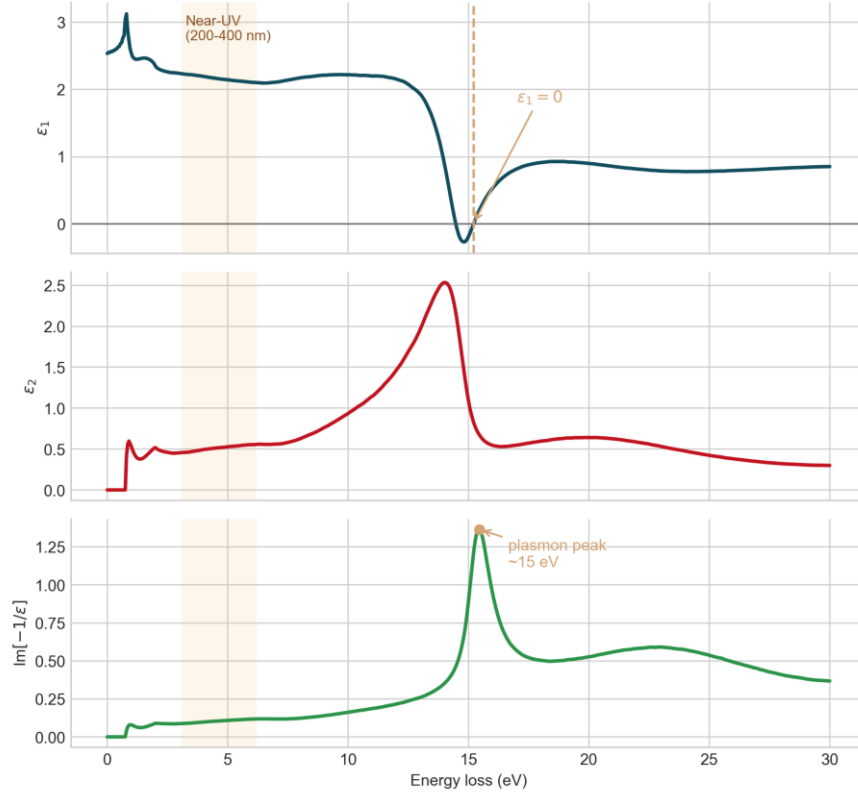


Fig. 9. Dielectric function reconstructed from STEM-EELS measurements. Top: real part of the permittivity ϵ_1 showing positive values across the near-UV region. Middle: imaginary part ϵ_2 indicating absorption in the UV. Bottom: energy-loss function $\text{Im}(-1/\epsilon)$ showing a plasmon peak near ~ 15 eV.

4.5) Ultraviolet Optical Response (UV–Vis Reflectance)

The optical response of the Al/Al₂O₃ multilayer system was characterized using UV–visible reflectance spectroscopy, with the normalized reflectance spectra shown in Fig. 10. These measurements provide insight into the wavelength-dependent interaction of light with the multilayer and serve as a macroscopic probe of the dielectric behavior discussed in Section 4.4.

The analysis is centered on the ultraviolet region (200–400 nm), which is the primary spectral range of interest in this study. Within this region, both the multilayer sample and the substrate exhibit relatively high reflectance values, with only moderate variation across wavelength. The multilayer consistently shows a reduced reflectance compared to the substrate, indicating increased absorption within the structure.

This reduction in reflectance can be directly linked to the dielectric response identified in the STEM-EELS analysis. As shown in Section 4.4, the ultraviolet regime is characterized by positive ϵ_1 and non-negligible ϵ_2 , corresponding to a lossy optical response. The presence of a finite imaginary permittivity leads to absorption of incident light, which manifests in the reflectance spectrum as a decrease relative to the substrate.

Importantly, no sharp features or pronounced resonant dips are observed in the UV range. Instead, the reflectance varies smoothly with wavelength, suggesting that the optical response is governed primarily by absorption and scattering processes rather than by strongly localized resonant modes.

At longer wavelengths, outside the primary UV region, more pronounced variations in reflectance are observed, including a broad minimum in the visible range. While these features indicate additional optical processes, they are not central to the ultraviolet response considered in this work.

Taken together, the UV–Vis results show that the multilayer exhibits a wavelength-dependent optical response in the ultraviolet region that is consistent with the dielectric behavior obtained from STEM-EELS, with absorption playing a dominant role in shaping the observed spectra.

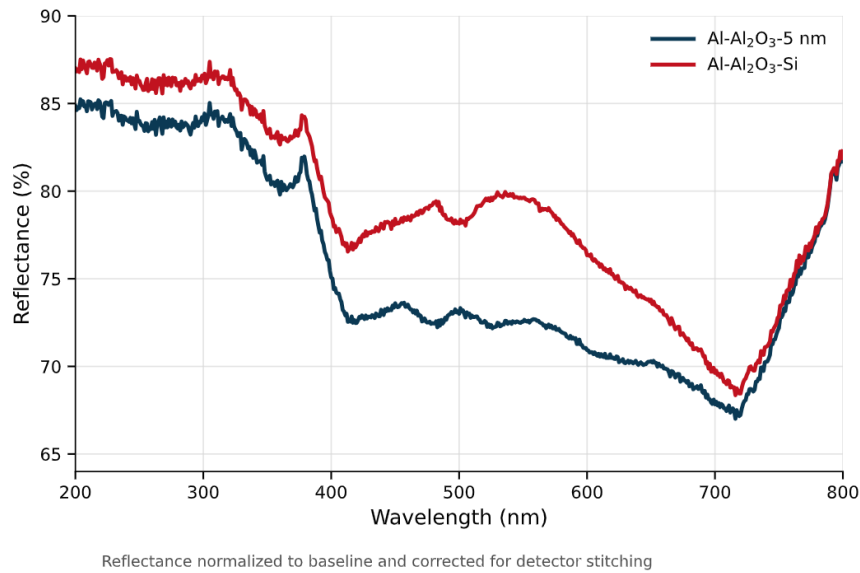


Fig. 10. Normalized reflectance spectra of the Al/Al₂O₃ multilayer and substrate across the UV–visible range. The multilayer exhibits reduced reflectance in the ultraviolet region, indicating increased absorption consistent with the dielectric response.

4.6) Cathodoluminescence Emission Characteristics

The optical emission properties of the Al/Al₂O₃ multilayer system were investigated using cathodoluminescence (CL) spectroscopy, with the measured emission spectrum shown in Fig. 11. This technique probes radiative recombination processes under electron-beam excitation and provides insight into the electronic states and defect-related emission within the material.

The spectrum exhibits a broad emission peak centered near $\sim 2.5\text{--}2.7$ eV, corresponding to the visible range. This feature indicates the presence of dominant radiative recombination processes, which are commonly associated with defect states and localized electronic transitions in oxide

materials. In particular, aluminum oxide is known to host oxygen vacancies and interface-related states that can contribute to luminescence in this energy range.

In addition to the main emission peak, the spectrum shows a gradual extension toward higher energies, reaching into the near-ultraviolet region (up to ~ 4 eV). While the intensity in this region is lower compared to the visible peak, its presence suggests that higher-energy radiative channels are also active within the multilayer system. These contributions are consistent with the broad distribution of electronic states arising from the nanoscale structure and compositional variations identified in Sections 4.2 and 4.3.

The overall spectral profile is smooth and continuous, with no sharp emission lines or discrete peaks. This behavior indicates that the emission is not dominated by well-defined electronic transitions, but rather by a distribution of localized states. Such a response is typical of systems with structural and compositional inhomogeneity, where variations at the interfaces give rise to a range of recombination pathways.

From the perspective of this study, the CL results provide complementary information to the dielectric and reflectance measurements. While the dominant emission occurs in the visible range, the extension into the near-UV confirms that the multilayer supports electronic transitions across a broad energy range. These observations are consistent with the absorption-dominated optical response identified in Section 4.5 and reflect the role of defect states and interface characteristics in shaping the overall optical behavior of the system.

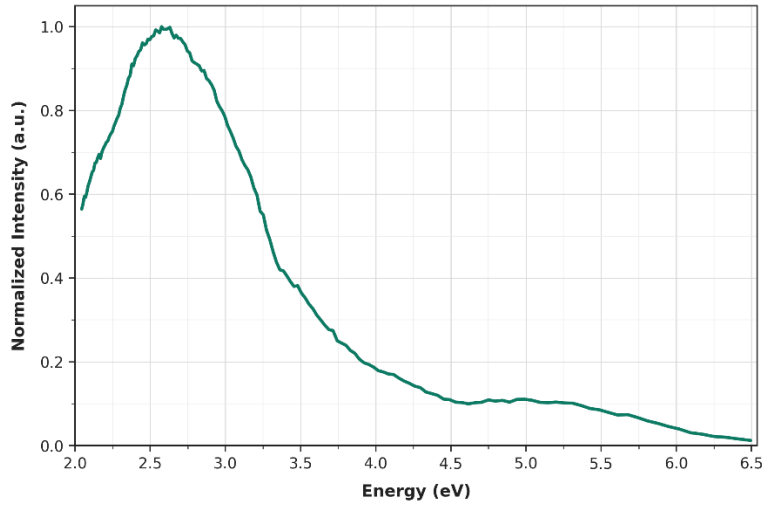


Fig. 11. Cathodoluminescence emission spectrum of the Al/Al₂O₃ multilayer system, showing a broad peak near ~ 2.5 – 2.7 eV and an extended tail toward higher energies reaching into the near-UV region.

4.7) Photoluminescence Behavior in the Near-UV Region

The photoluminescence (PL) response of the Al/Al₂O₃ multilayer system was investigated to probe

radiative recombination processes in the near-ultraviolet region, with the emission spectrum shown in Fig. 12. The data correspond to the baseline-subtracted, off-band PL signal, with the excitation-dominated region removed to isolate intrinsic emission features.

The analysis is centered on the near-UV region around ~ 300 nm. In this range, a distinct emission feature is observed near ~ 310 – 320 nm, indicating the presence of radiative recombination channels at near-ultraviolet energies. This emission is consistent with defect-related transitions, likely associated with oxygen vacancies and interface states within the Al_2O_3 layers. These localized states introduce energy levels within the bandgap that enable recombination at energies corresponding to the near-UV range.

The presence of this emission feature aligns with the structural and compositional characteristics identified in earlier sections. As shown in Sections 4.2 and 4.3, the multilayer exhibits interface roughness and compositional broadening, which contribute to the formation of defect states. These states provide pathways for radiative recombination and are reflected in the observed PL signal.

Additional emission features are observed at longer wavelengths, including a dominant peak in the visible region. While these features indicate a broader distribution of defect-related states, they are not central to the ultraviolet response considered in this work. The focus on the near-UV region highlights the presence of higher-energy recombination processes that are directly relevant to the optical behavior of the multilayer at shorter wavelengths.

Overall, the PL results confirm that the multilayer system supports radiative recombination in the near-ultraviolet region, with emission features that are consistent with defect-mediated processes and interface-related states.

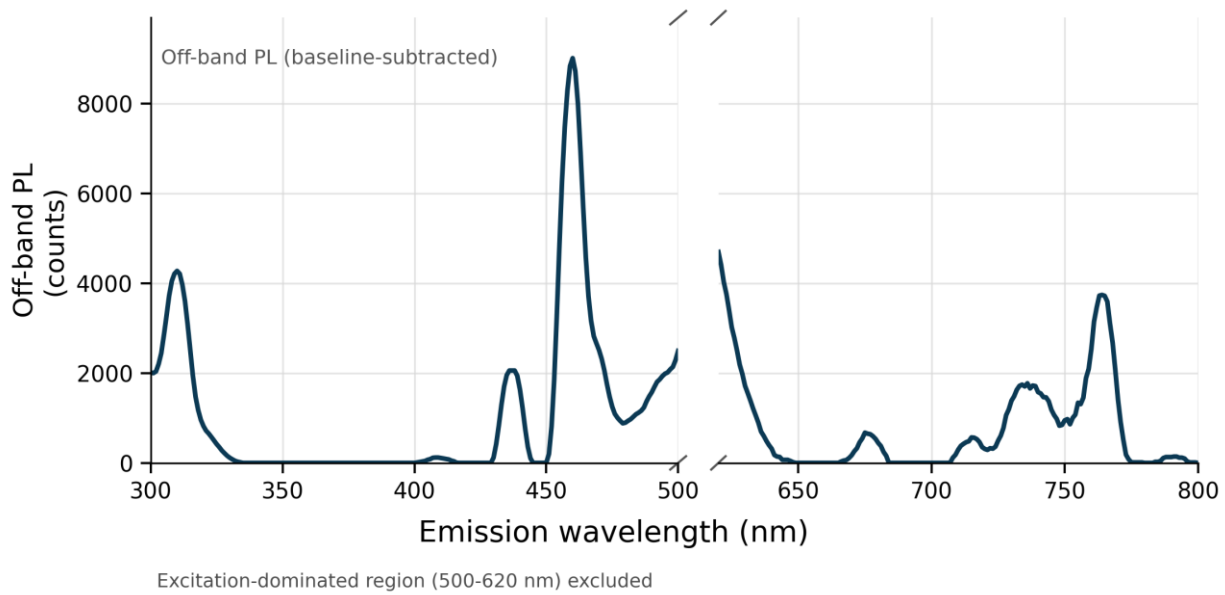


Fig. 12. Photoluminescence (PL) emission spectrum of the $\text{Al}/\text{Al}_2\text{O}_3$ multilayer system. The plot shows the baseline-subtracted off-band PL signal, with the excitation-dominated region (500–620

nm) excluded. A near-UV emission feature is observed around $\sim 310\text{--}320$ nm, along with additional peaks at longer wavelengths associated with defect-related recombination.

4.8) Discussion

The combined results from electromagnetic modeling, structural characterization, compositional analysis, dielectric reconstruction, and optical measurements provide a coherent picture of the behavior of the Al/Al₂O₃ multilayer system in the ultraviolet spectral range.

The FDTD simulations demonstrate that, under idealized conditions, the multilayer geometry supports directional wave propagation consistent with negative refraction. This behavior arises from the strong contrast between the metallic and dielectric layers, which enables an anisotropic electromagnetic response. The simulation therefore establishes a theoretical framework for understanding how multilayer structures can be engineered to produce metamaterial-like optical effects.

Experimentally, the dielectric response obtained from STEM-EELS confirms that the system supports plasmonic activity at higher energies, as evidenced by the pronounced energy-loss peak near ~ 15 eV. This indicates strong coupling between electromagnetic fields and collective electron oscillations in the aluminum layers. Within the ultraviolet range, the dielectric function exhibits a well-defined response characterized by positive ϵ_1 and finite ϵ_2 , which governs the interaction of light with the multilayer.

The UV–Vis reflectance measurements further support this interpretation. The reduced reflectance observed in the multilayer relative to the substrate indicates enhanced absorption, consistent with the non-negligible ϵ_2 obtained from dielectric reconstruction. The smooth spectral variation in the UV range suggests that the optical response is dominated by absorption and scattering processes rather than sharply defined resonant modes.

The structural and compositional analyses provide important context for these observations. TEM and HAADF-STEM imaging reveal a continuous multilayer structure with realistic nanoscale morphology, including interface waviness and thickness variations. EDS mapping shows periodic elemental modulation with gradual transitions at the interfaces, indicating compositional broadening. Together, these features contribute to a distributed electronic environment that influences the dielectric response and optical behavior of the system.

The cathodoluminescence and photoluminescence measurements further highlight the role of defect-related and interface-mediated processes. Broad emission features and multiple recombination channels indicate the presence of localized states within the multilayer, which are consistent with the structural and compositional characteristics observed. The detection of emission extending into the near-ultraviolet region confirms that higher-energy electronic transitions are supported within the system.

Overall, the results demonstrate that the Al/Al₂O₃ multilayer exhibits a well-defined

electromagnetic and optical response in the ultraviolet region, governed by the interplay between material properties, structural characteristics, and nanoscale composition. The agreement between simulation and experiment, when considered within the context of realistic material behavior, provides a consistent understanding of the system and highlights the key factors that shape its response.

5) Future Work

The results presented in this study provide a comprehensive understanding of the electromagnetic, structural, and optical behavior of the Al/Al₂O₃ multilayer system in the ultraviolet spectral range. Building on these findings, several directions for future work can be identified to further refine the system and enhance its optical performance.

One key area for improvement is the structural uniformity of the multilayer. As demonstrated in the structural and compositional analyses, the presence of interface waviness and compositional broadening introduces variations in the local dielectric environment. Future efforts should therefore focus on optimizing the fabrication process to achieve smoother interfaces and more uniform layer thickness. Improving interface quality is expected to enhance the consistency of the metal–dielectric contrast and provide a more controlled electromagnetic response.

In addition to structural refinement, further optimization of the multilayer geometry is an important direction. Adjusting layer thicknesses and improving symmetry between the metal and dielectric layers may enable more controlled tuning of the effective optical properties. Such modifications can enhance the degree of anisotropy and improve the interaction between electromagnetic fields and the multilayer structure.

Another important consideration is the reduction of optical losses. The dielectric reconstruction and optical measurements indicate that absorption plays a significant role in the ultraviolet response of the system. Future work may explore strategies to minimize these losses, such as refining material quality, reducing defect density, or investigating alternative deposition conditions. Reducing losses would enable stronger light–matter interaction and improve the clarity of the optical response.

To further improve the accuracy of the dielectric characterization, additional experimental techniques can be incorporated. In particular, spectroscopic ellipsometry would provide an independent measurement of the complex dielectric function, enabling more precise determination of ϵ_1 and ϵ_2 across the relevant spectral range. This would complement the STEM-EELS analysis and provide a more complete description of the optical properties of the multilayer.

Finally, the electromagnetic modeling can be refined by incorporating experimentally derived parameters. Using dielectric functions obtained directly from measurements, along with realistic structural features such as interface roughness and thickness variation, would improve the

agreement between simulation and experiment. This approach would allow for a more accurate representation of the system and provide deeper insight into the relationship between structure and optical response.

Overall, these directions provide a clear pathway for advancing the understanding and performance of multilayer systems for ultraviolet photonic applications.

6) Conclusion

This study investigated the electromagnetic, structural, dielectric, and optical behavior of Al/Al₂O₃ multilayer systems with a focus on their response in the ultraviolet spectral range. Through a combination of numerical simulation and experimental characterization, a consistent picture of the system's behavior has been established.

The FDTD simulations demonstrate that multilayer metal–dielectric structures can support directional wave propagation consistent with negative refraction under idealized conditions. This behavior arises from the strong contrast between metallic and dielectric layers and highlights the potential of such systems for engineered optical responses.

Experimentally, the dielectric and optical measurements confirm that the multilayer exhibits a well-defined response in the ultraviolet region. The STEM-EELS analysis reveals a frequency-dependent dielectric function with a plasmon resonance at higher energies, while the UV–Vis measurements show a wavelength-dependent reflectance dominated by absorption. Together, these results indicate that the optical response in the UV range is governed by the intrinsic material properties of the system.

The structural and compositional analyses further demonstrate that the fabricated multilayer exhibits realistic nanoscale morphology, including interface roughness and compositional variation. These features influence the local dielectric environment and contribute to the overall optical behavior. The cathodoluminescence and photoluminescence measurements provide additional evidence of defect-related and interface-mediated processes, confirming the role of localized electronic states in shaping the emission characteristics.

Taken together, the results establish a clear relationship between structure, composition, dielectric response, and electromagnetic behavior in Al/Al₂O₃ multilayers. This framework provides a foundation for understanding how nanoscale features influence optical properties and supports the continued development of multilayer systems for ultraviolet photonic applications.

References

- [1] V. G. Veselago, “The electrodynamics of substances with simultaneously negative values of ϵ and μ ,” *Sov. Phys. Usp.*, vol. 10, no. 4, pp. 509–514, Jan.–Feb. 1968, doi: 10.1070/PU1968v010n04ABEH003699.
- [2] J. B. Pendry, “Negative refraction makes a perfect lens,” *Phys. Rev. Lett.*, vol. 85, no. 18, pp. 3966–3969, Oct. 2000, doi: 10.1103/PhysRevLett.85.3966.
- [3] N. Engheta and R. W. Ziolkowski, *Metamaterials: Physics and Engineering Explorations*. Hoboken, NJ, USA: Wiley, 2006.
- [4] A. Poddubny, I. Iorsh, P. Belov, and Y. Kivshar, “Hyperbolic metamaterials,” *Nat. Photon.*, vol. 7, no. 12, pp. 948–957, Dec. 2013, doi: 10.1038/nphoton.2013.243.
- [5] Z. Jacob, L. V. Alekseyev, and E. Narimanov, “Optical hyperlens: Far-field imaging beyond the diffraction limit,” *Opt. Express*, vol. 14, no. 18, pp. 8247–8256, Sep. 2006, doi: 10.1364/OE.14.008247.
- [6] D. R. Smith and J. B. Pendry, “Homogenization of metamaterials by field averaging,” *J. Opt. Soc. Am. B*, vol. 23, no. 3, pp. 391–403, Mar. 2006, doi: 10.1364/JOSAB.23.000391.
- [7] I. I. Smolyaninov, Y.-J. Hung, and C. C. Davis, “Magnifying superlens in the visible frequency range,” *Science*, vol. 315, no. 5819, pp. 1699–1701, Mar. 2007, doi: 10.1126/science.1139848.
- [8] P. B. Johnson and R. W. Christy, “Optical constants of the noble metals,” *Phys. Rev. B*, vol. 6, no. 12, pp. 4370–4379, Dec. 1972, doi: 10.1103/PhysRevB.6.4370.
- [9] E. D. Palik, *Handbook of Optical Constants of Solids*. San Diego, CA, USA: Academic Press, 1998.
- [10] M. W. Knight, N. S. King, L. Liu, H. O. Everitt, P. Nordlander, and N. J. Halas, “Aluminum for plasmonics,” *ACS Nano*, vol. 8, no. 1, pp. 834–840, Jan. 2014, doi: 10.1021/nn405584b.
- [11] S. A. Maier, *Plasmonics: Fundamentals and Applications*. New York, NY, USA: Springer, 2007.
- [12] J. M. McMahon, S. K. Gray, and G. C. Schatz, “Optical properties of nanostructured aluminum for plasmonics,” *Chem. Rev.*, vol. 111, no. 6, pp. 3737–3758, Jun. 2011, doi: 10.1021/cr100302a.

- [13] Murata Manufacturing Co., “What is a metamaterial?” Murata Manufacturing Co. [Online]. Available: <https://article.murata.com/en-global/article/what-is-a-metamaterial>.
- [14] N. W. Ashcroft and N. D. Mermin, *Solid State Physics*. New York, NY, USA: Holt, Rinehart and Winston, 1976.
- [15] P. Yeh, *Optical Waves in Layered Media*. New York, NY, USA: Wiley, 1988.
- [16] D. R. Smith and D. Schurig, “Electromagnetic wave propagation in media with indefinite permittivity and permeability tensors,” *Phys. Rev. Lett.*, vol. 90, no. 7, p. 077405, Feb. 2003, doi: 10.1103/PhysRevLett.90.077405.
- [17] A. Boltasseva and H. A. Atwater, “Low-loss plasmonic metamaterials,” *Science*, vol. 331, no. 6015, pp. 290–291, Jan. 2011, doi: 10.1126/science.1198258.
- [18] C. M. Soukoulis and M. Wegener, “Past achievements and future challenges in the development of three-dimensional photonic metamaterials,” *Nat. Photon.*, vol. 5, no. 9, pp. 523–530, Sep. 2011, doi: 10.1038/nphoton.2011.154.
- [19] F. Wooten, *Optical Properties of Solids*. New York, NY, USA: Academic Press, 1972.
- [20] H. Raether, *Surface Plasmons on Smooth and Rough Surfaces and on Gratings*. Berlin, Germany: Springer, 1988.
- [21] R. F. Egerton, *Electron Energy-Loss Spectroscopy in the Electron Microscope*, 3rd ed. New York, NY, USA: Springer, 2011.
- [22] F. J. García de Abajo, “Optical excitations in electron microscopy,” *Rev. Mod. Phys.*, vol. 82, no. 1, pp. 209–275, Jan. 2010, doi: 10.1103/RevModPhys.82.209.
- [23] A. D. Rakić, A. B. Djurišić, J. M. Elazar, and M. L. Majewski, “Optical properties of metallic films for vertical-cavity optoelectronic devices,” *Appl. Opt.*, vol. 37, no. 22, pp. 5271–5283, Aug. 1998, doi: 10.1364/AO.37.005271.
- [24] H. H. Tippins, “Optical absorption and photoconductivity in the band edge of $\alpha\text{-Al}_2\text{O}_3$,” *Phys. Rev.*, vol. 140, no. 1A, pp. A316–A319, Oct. 1965, doi: 10.1103/PhysRev.140.A316.
- [25] M. A. Stevens-Kalceff, “Cathodoluminescence microcharacterization of point defects in $\alpha\text{-Al}_2\text{O}_3$,” *J. Phys. D: Appl. Phys.*, vol. 42, no. 21, Art. no. 213001, Nov. 2009, doi: 10.1088/0022-3727/42/21/213001.

- [26] J. Zhao, J. Gao, Y. Deng, H. Liu, and X. Wang, “Negative refraction by a planar Ag/SiO₂ multilayer at ultraviolet wavelength to the limit of silver,” *AIP Adv.*, vol. 4, no. 4, Art. no. 047128, Apr. 2014, doi: 10.1063/1.4873156.
- [27] P. Zheng, “Plasmon-enhanced optical sensing by engineering metallic nanostructures,” Ph.D. dissertation, Benjamin M. Statler College of Eng. and Mineral Resources, West Virginia Univ., Morgantown, WV, USA, 2018.
- [28] A. Taflove and S. C. Hagness, *Computational Electrodynamics: The Finite-Difference Time-Domain Method*, 3rd ed. Norwood, MA, USA: Artech House, 2005.
- [29] K. S. Yee, “Numerical solution of initial boundary value problems involving Maxwell’s equations in isotropic media,” *IEEE Trans. Antennas Propag.*, vol. 14, no. 3, pp. 302–307, May 1966, doi: 10.1109/TAP.1966.1138693.
- [30] J.-P. Berenger, “A perfectly matched layer for the absorption of electromagnetic waves,” *J. Comput. Phys.*, vol. 114, no. 2, pp. 185–200, Oct. 1994, doi: 10.1006/jcph.1994.1159.
- [31] D. M. Sullivan, *Electromagnetic Simulation Using the FDTD Method*, 2nd ed. Hoboken, NJ, USA: Wiley-IEEE Press, 2013.
- [32] J. B. Pendry and D. R. Smith, “Reversing light with negative refraction,” *Phys. Today*, vol. 57, no. 6, pp. 37–43, Jun. 2004, doi: 10.1063/1.1784272.
- [33] V. M. Shalaev, “Optical negative-index metamaterials,” *Nat. Photon.*, vol. 1, no. 1, pp. 41–48, Jan. 2007, doi: 10.1038/nphoton.2006.49.
- [34] D. M. Mattox, *Handbook of Physical Vapor Deposition (PVD) Processing*, 2nd ed. Oxford, U.K.: William Andrew, 2010.
- [35] M. Ohring, *Materials Science of Thin Films*, 2nd ed. San Diego, CA, USA: Academic Press, 2002.
- [36] L. P. H. Jeurgens, W. G. Sloof, F. D. Tichelaar, and E. J. Mittemeijer, “Growth kinetics and mechanisms of aluminum-oxide films formed by thermal oxidation of aluminum,” *J. Appl. Phys.*, vol. 92, no. 3, pp. 1649–1656, Aug. 2002, doi: 10.1063/1.1491591.
- [37] D. B. Williams and C. B. Carter, *Transmission Electron Microscopy: A Textbook for Materials Science*, 2nd ed. New York, NY, USA: Springer, 2009.
- [38] L. A. Giannuzzi and F. A. Stevie, *Introduction to Focused Ion Beams: Instrumentation, Theory, Techniques and Practice*. New York, NY, USA: Springer, 2005.

- [39] S. J. Pennycook and P. D. Nellist, *Scanning Transmission Electron Microscopy: Imaging and Analysis*. New York, NY, USA: Springer, 2011.
- [40] J. I. Goldstein et al., *Scanning Electron Microscopy and X-Ray Microanalysis*, 4th ed. New York, NY, USA: Springer, 2018.
- [41] J. I. Pankove, *Optical Processes in Semiconductors*. New York, NY, USA: Dover, 1971.
- [42] B. G. Yacobi and D. B. Holt, *Cathodoluminescence Microscopy of Inorganic Solids*. New York, NY, USA: Plenum Press, 1990.

Supplement

Python Code for analyzing STEM EELS data

```
from pathlib import Path
import matplotlib
import numpy as np

matplotlib.use("Agg")
import matplotlib.pyplot as plt

ROOT = Path(__file__).resolve().parent

def load_curve(filename: str) -> tuple[np.ndarray, np.ndarray]:
    data = np.loadtxt(ROOT / filename, delimiter=",")
    return data[:, 0], data[:, 1]

def main() -> None:
    energy_eps1, eps1 = load_curve("Epsilon 1.csv")
    energy_eps2, eps2 = load_curve("Epsilon 2.csv")
    energy_loss, loss = load_curve("Energy-loss function.csv")

    mask = (energy_eps1 >= 0) & (energy_eps1 <= 30)
    mask_loss = (energy_loss >= 0) & (energy_loss <= 30)

    plt.style.use("seaborn-v0_8-whitegrid")
    fig, axes = plt.subplots(
        3,
        1,
        figsize=(9, 8.5),
        sharex=True,
        gridspec_kw={"height_ratios": [1.1, 1.0, 1.0], "hspace": 0.08},
    )

    colors = {
        "eps1": "#0f4c5c",
        "eps2": "#c1121f",
        "loss": "#2b9348",
        "accent": "#d4a373",
    }

    axes[0].plot(energy_eps1[mask], eps1[mask], color=colors["eps1"], linewidth=2.2)
```

```

axes[0].axhline(0, color="#555555", linewidth=1, alpha=0.8)
axes[0].set_ylabel(r"$\varepsilon_1$")
axes[0].set_title("Dielectric Reconstruction and Energy-Loss Response (EELS)", pad=12)

sign = np.sign(eps1[mask])
crossings = np.where(np.diff(sign) != 0)[0]
if len(crossings):
    idx = crossings[np.argmin(np.abs(energy_eps1[mask][crossings] - 15))]
    x0 = energy_eps1[mask][idx]
    x1 = energy_eps1[mask][idx + 1]
    y0 = eps1[mask][idx]
    y1 = eps1[mask][idx + 1]
    crossing_energy = x0 - y0 * (x1 - x0) / (y1 - y0)
    axes[0].axvline(crossing_energy, color=colors["accent"], linestyle="--", linewidth=1.5)
    axes[0].annotate(
        r"$\varepsilon_1 = 0$",
        xy=(crossing_energy, 0),
        xytext=(crossing_energy + 1.8, np.nanmax(eps1[mask]) * 0.58),
        arrowprops={"arrowstyle": "->", "color": colors["accent"], "lw": 1.1},
        color=colors["accent"],
        fontsize=10,
    )

axes[1].plot(energy_eps2[mask], eps2[mask], color=colors["eps2"], linewidth=2.2)
axes[1].set_ylabel(r"$\varepsilon_2$")

peak_idx = np.argmax(loss[mask_loss])
peak_energy = energy_loss[mask_loss][peak_idx]
peak_value = loss[mask_loss][peak_idx]

axes[2].plot(energy_loss[mask_loss], loss[mask_loss], color=colors["loss"], linewidth=2.2)
axes[2].scatter([peak_energy], [peak_value], color=colors["accent"], s=35, zorder=3)
axes[2].annotate(
    "plasmon peak\n~15 eV",
    xy=(peak_energy, peak_value),
    xytext=(peak_energy + 1.2, peak_value * 0.82),
    arrowprops={"arrowstyle": "->", "color": colors["accent"], "lw": 1.1},
    color=colors["accent"],
    fontsize=10,
)
axes[2].set_ylabel(r"$\text{Im}[-1/\varepsilon]$")
axes[2].set_xlabel("Energy loss (eV)")

uv_start = 3.1
uv_end = 6.2
for ax in axes:

```

```

    ax.axvspan(uv_start, uv_end, color="#f6bd60", alpha=0.12, linewidth=0)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)

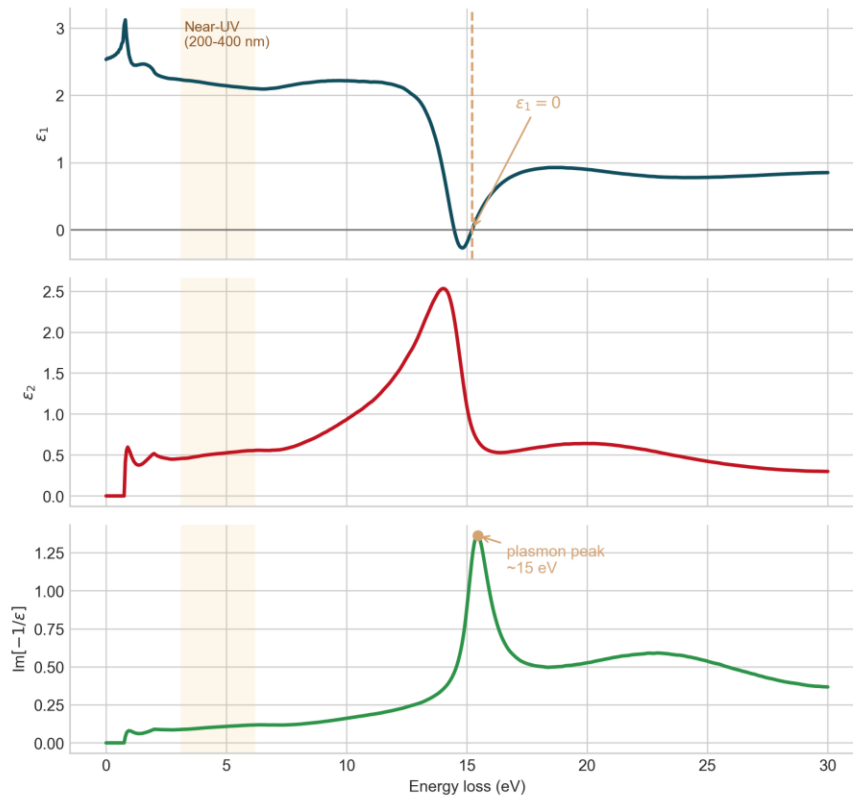
    axes[0].text(
        uv_start + 0.15,
        axes[0].get_ylim()[1] * 0.83,
        "Near-UV\n(200-400 nm)",
        color="#8d5524",
        fontsize=9,
    )

    fig.text(
        0.015,
        0.015,
        "Story for slide: measured energy-loss response -> reconstructed dielectric function ->
plasmon behavior",
        fontsize=9,
        color="#444444",
    )

    output = ROOT / "EELS_summary_combined.png"
    fig.savefig(output, dpi=300, bbox_inches="tight")
    plt.close(fig)
    print(output)

if __name__ == "__main__":
    main()

```



Python Code for analyzing UV-Vis data

```
from __future__ import annotations

import csv
from pathlib import Path

import matplotlib
import numpy as np

matplotlib.use("Agg")
import matplotlib.pyplot as plt

ROOT = Path(__file__).resolve().parent
SCAN_DIR = ROOT / "Scan - Lambda 1050 Friday, January 23, 2026 10_47 AM Arabian Standard Time"
OUT_DIR = ROOT / "publishable"

BOUNDARIES = (379.5, 728.5)
FIT_WINDOWS = {
    379.5: ((360.0, 378.0), (381.0, 399.0)),
    728.5: ((710.0, 728.0), (729.0, 747.0)),
}
```

```

def read_percent_reflectance(filename: str) -> tuple[np.ndarray, np.ndarray]:
    path = SCAN_DIR / filename
    with path.open(newline="") as handle:
        reader = csv.reader(handle, skipinitialspace=True)
        next(reader)
        rows = [(float(row[0]), float(row[1])) for row in reader if len(row) >= 2]
    data = np.array(sorted(rows), dtype=float)
    return data[:, 0], data[:, 1]

def fit_at_boundary(wavelength: np.ndarray, values: np.ndarray, boundary: float) -> tuple[float, float]:
    left_window, right_window = FIT_WINDOWS[boundary]
    left_mask = (wavelength >= left_window[0]) & (wavelength <= left_window[1])
    right_mask = (wavelength >= right_window[0]) & (wavelength <= right_window[1])
    left_fit = np.polyfit(wavelength[left_mask], values[left_mask], 1)
    right_fit = np.polyfit(wavelength[right_mask], values[right_mask], 1)
    return np.polyval(left_fit, boundary), np.polyval(right_fit, boundary)

def normalize_and_stitch(
    wavelength: np.ndarray, raw_percent_r: np.ndarray, baseline_percent_r: np.ndarray
) -> tuple[np.ndarray, np.ndarray, dict[str, float]]:
    normalized = raw_percent_r / baseline_percent_r * 100.0
    cleaned = normalized.copy()

    uv_pred, visible_pred = fit_at_boundary(wavelength, normalized, 379.5)
    visible_pred_left, nir_pred = fit_at_boundary(wavelength, normalized, 728.5)

    uv_scale = visible_pred / uv_pred
    nir_scale = visible_pred_left / nir_pred

    cleaned[wavelength <= BOUNDARIES[0]] *= uv_scale
    cleaned[wavelength >= BOUNDARIES[1]] *= nir_scale

    meta = {
        "uv_scale": uv_scale,
        "nir_scale": nir_scale,
        "uv_boundary_nm": BOUNDARIES[0],
        "nir_boundary_nm": BOUNDARIES[1],
    }
    return normalized, cleaned, meta

def save_cleaned_csv(
    wavelength: np.ndarray,

```

```

baseline: np.ndarray,
raw_curves: dict[str, np.ndarray],
normalized_curves: dict[str, np.ndarray],
cleaned_curves: dict[str, np.ndarray],
) -> Path:
    OUT_DIR.mkdir(exist_ok=True)
    output = OUT_DIR / "uvvis_cleaned_reflectance.csv"
    with output.open("w", newline="") as handle:
        writer = csv.writer(handle)
        writer.writerow(
            [
                "wavelength_nm",
                "baseline_percent_reflectance",
                "S1_raw_percent_reflectance",
                "S2_raw_percent_reflectance",
                "S1_baseline_normalized_percent_reflectance",
                "S2_baseline_normalized_percent_reflectance",
                "S1_cleaned_percent_reflectance",
                "S2_cleaned_percent_reflectance",
            ]
        )
        for i, nm in enumerate(wavelength):
            writer.writerow(
                [
                    f"{nm:.1f}",
                    f"{baseline[i]:.6f}",
                    f"{raw_curves['S1'][i]:.6f}",
                    f"{raw_curves['S2'][i]:.6f}",
                    f"{normalized_curves['S1'][i]:.6f}",
                    f"{normalized_curves['S2'][i]:.6f}",
                    f"{cleaned_curves['S1'][i]:.6f}",
                    f"{cleaned_curves['S2'][i]:.6f}",
                ]
            )
    return output

def save_metadata(scales: dict[str, dict[str, float]]) -> Path:
    output = OUT_DIR / "uvvis_cleaning_notes.txt"
    lines = [
        "UV-Vis cleaning workflow",
        "1. Baseline-normalized reflectance: sample / baseline * 100",
        "2. Corrected detector-switch discontinuities by piecewise scaling",
        "3. Middle segment (380-728 nm) used as the anchor region",
        "",
    ]

```

```

    for label, meta in scales.items():
        lines.extend(
            [
                f"{label}",
                f"    UV-segment scale (<= {meta['uv_boundary_nm']:.1f} nm):
{meta['uv_scale']:.6f}",
                f"    NIR-segment scale (>= {meta['nir_boundary_nm']:.1f} nm):
{meta['nir_scale']:.6f}",
                "",
            ]
        )
    output.write_text("\n".join(lines), encoding="ascii")
    return output

def style_axes(ax: plt.Axes, ylabel: str = "Reflectance (%)") -> None:
    ax.set_xlabel("Wavelength (nm)")
    ax.set_ylabel(ylabel)
    ax.grid(True, color="#d9d9d9", linewidth=0.8, alpha=0.7)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.tick_params(direction="out", length=4.5, width=1)

def make_plot(
    wavelength: np.ndarray,
    curves: dict[str, np.ndarray],
    stem: str,
    note: str | None = None,
    ylim: tuple[float, float] | None = None,
) -> list[Path]:
    plt.rcParams.update(
        {
            "font.size": 11,
            "axes.labelsize": 12,
            "xtick.labelsize": 10,
            "ytick.labelsize": 10,
            "legend.fontsize": 10,
        }
    )
    fig, ax = plt.subplots(figsize=(7.2, 4.8))
    fig.patch.set_facecolor("white")
    ax.set_facecolor("white")

    colors = {
        "A1-A1$ 2$0$ 3$-5 nm": "#0b3954",
    }

```

```

    "Al-Al$_2$O$_3$-Si": "#c1121f",
}

for label, values in curves.items():
    ax.plot(wavelength, values, linewidth=2.3, color=colors[label], label=label)

ax.set_xlim(200, 800)
if ylim is None:
    ymin = min(np.min(v) for v in curves.values())
    ymax = max(np.max(v) for v in curves.values())
    ylim = (np.floor((ymin - 1.5) / 2) * 2, min(102, np.ceil((ymax + 1.0) / 2) * 2))
ax.set_ylim(*ylim)
ax.set_yticks(np.arange(np.ceil(ylim[0] / 5) * 5, ylim[1] + 0.1, 5))
ax.legend(frameon=False, loc="upper right", handlelength=2.4)
style_axes(ax)

if note:
    fig.text(0.125, 0.02, note, ha="left", va="bottom", fontsize=9, color="#4d4d4d")

plt.tight_layout(rect=(0, 0.05 if note else 0, 1, 1))

outputs = []
for suffix in ("png", "svg", "pdf"):
    target = OUT_DIR / f"{stem}.{suffix}"
    fig.savefig(target, dpi=300, bbox_inches="tight")
    outputs.append(target)
plt.close(fig)
return outputs

def main() -> None:
    wavelength, baseline = read_percent_reflectance("100% or 0 Absorbance
Baseline.Correction.Raw.csv")
    wavelength_s1, raw_s1 = read_percent_reflectance("S1.Sample.Raw.csv")
    wavelength_s2, raw_s2 = read_percent_reflectance("S2.Sample.Raw.csv")

    if not (np.array_equal(wavelength, wavelength_s1) and np.array_equal(wavelength,
wavelength_s2)):
        raise ValueError("Input scans do not share the same wavelength grid.")

    norm_s1, cleaned_s1, meta_s1 = normalize_and_stitch(wavelength, raw_s1, baseline)
    norm_s2, cleaned_s2, meta_s2 = normalize_and_stitch(wavelength, raw_s2, baseline)

    save_cleaned_csv(
        wavelength,
        baseline,

```

```

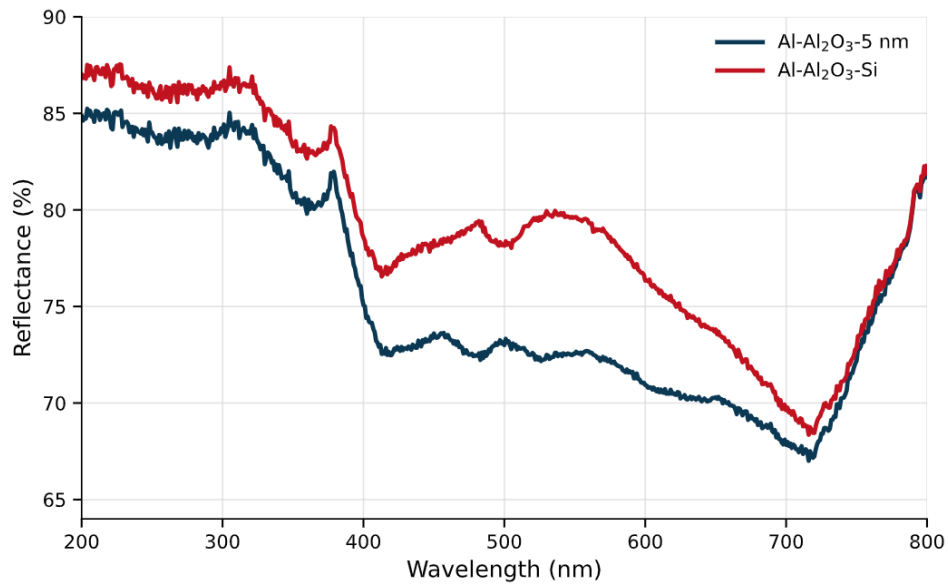
        {"S1": raw_s1, "S2": raw_s2},
        {"S1": norm_s1, "S2": norm_s2},
        {"S1": cleaned_s1, "S2": cleaned_s2},
    )
    save_metadata({"S1 (Al-Al2O3-5 nm)": meta_s1, "S2 (Al-Al2O3-Si)": meta_s2})
    corrected_outputs = make_plot(
        wavelength,
        {
            "Al-Al$_2$O$_3$-5 nm": cleaned_s1,
            "Al-Al$_2$O$_3$-Si": cleaned_s2,
        },
        stem="uvvis_reflectance_corrected_publication_v2",
        note="Reflectance normalized to baseline and corrected for detector stitching",
        ylim=(64, 90),
    )

    raw_outputs = make_plot(
        wavelength,
        {
            "Al-Al$_2$O$_3$-5 nm": raw_s1,
            "Al-Al$_2$O$_3$-Si": raw_s2,
        },
        stem="uvvis_reflectance_raw_publication",
        ylim=(80, 100),
    )

    for path in corrected_outputs + raw_outputs:
        print(path)

if __name__ == "__main__":
    main()

```



Reflectance normalized to baseline and corrected for detector stitching

Python Code for analyzing CL data

```
from __future__ import annotations

from pathlib import Path

import matplotlib
import pandas as pd

matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.ticker import AutoMinorLocator, MultipleLocator

HC_EV_NM = 1239.841984
INPUT_FILE = Path(r"c:\Users\hp\Downloads\CL Spectrum_5kV_13nA_400 nm.msa")
OUTPUT_DIR = Path(r"c:\Users\hp\Downloads\Data Analysis\output")

def style_axes(ax, x_label: str, y_label: str, y_max: float | None = None) -> None:
    ax.set_facecolor("white")
    ax.set_xlabel(x_label, fontsize=11, fontweight="semibold", labelpad=8)
    ax.set_ylabel(y_label, fontsize=11, fontweight="semibold", labelpad=8)

    ax.tick_params(
        axis="both",
```

```

        which="major",
        direction="out",
        length=4,
        width=0.9,
        labelsiz=10,
        colors="#222222",
    )
ax.tick_params(
    axis="both",
    which="minor",
    direction="out",
    length=2.5,
    width=0.7,
    colors="#444444",
)

for side in ("top", "right", "bottom", "left"):
    ax.spines[side].set_linewidth(1.0)
    ax.spines[side].set_color("#222222")

ax.grid(True, which="major", color="#d9d9d9", linewidth=0.7, alpha=0.9)
ax.grid(True, which="minor", color="#e6e6e6", linewidth=0.5, alpha=0.9)

ax.xaxis.set_major_locator(MultipleLocator(0.5))
ax.xaxis.set_minor_locator(AutoMinorLocator(2))
ax.yaxis.set_minor_locator(AutoMinorLocator(2))
ax.margins(x=0.01)

if y_max is not None:
    ax.set_ylim(0, y_max)

def parse_msa(path: Path) -> tuple[dict[str, str], pd.DataFrame]:
    metadata: dict[str, str] = {}
    rows: list[tuple[float, float]] = []
    in_spectrum = False

    for raw_line in path.read_text().splitlines():
        line = raw_line.strip()
        if not line:
            continue
        if line.startswith("#"):
            if ":" in line:
                key, value = line.split(":", 1)
                metadata[key.strip()] = value.strip()
            if line.startswith("#SPECTRUM"):

```

```

        in_spectrum = True
    elif line.startswith("#ENDOFDATA"):
        in_spectrum = False
    continue
if in_spectrum and "," in line:
    x_nm, counts = [float(part.strip()) for part in line.split(",", 1)]
    rows.append((x_nm, counts))

df = pd.DataFrame(rows, columns=["wavelength_nm", "counts"])
return metadata, df

def main() -> None:
    metadata, df = parse_msa(INPUT_FILE)
    if df.empty:
        raise ValueError(f"No spectrum data found in {INPUT_FILE}")

    df["energy_eV"] = HC_EV_NM / df["wavelength_nm"]
    df = df.sort_values("energy_eV").reset_index(drop=True)
    df["normalized_counts"] = df["counts"] / df["counts"].max()

    OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
    csv_path = OUTPUT_DIR / "cl_spectrum_ev.csv"
    raw_plot_path = OUTPUT_DIR / "cl_spectrum_ev_counts.png"
    norm_plot_path = OUTPUT_DIR / "cl_spectrum_ev_normalized.png"

    df.to_csv(csv_path, index=False)

    matplotlib.rcParams.update(
        {
            "font.family": "DejaVu Sans",
            "axes.titleweight": "semibold",
            "axes.labelcolor": "#222222",
            "xtick.color": "#222222",
            "ytick.color": "#222222",
        }
    )

    fig, ax = plt.subplots(figsize=(7.2, 4.8), dpi=300)
    fig.patch.set_facecolor("white")
    ax.plot(
        df["energy_eV"],
        df["counts"],
        color="#1b4f9c",
        linewidth=2.1,
        solid_capstyle="round",

```

```

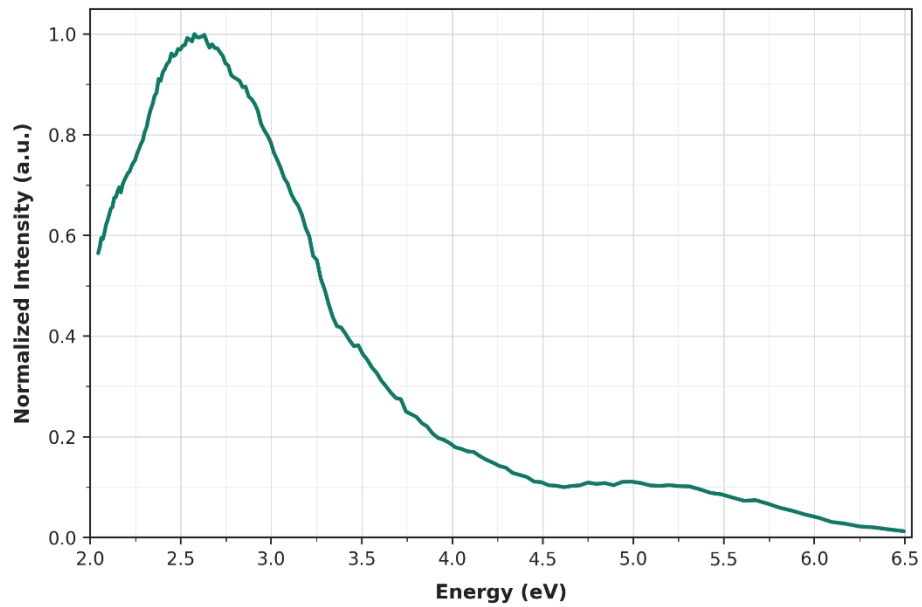
)
style_axes(ax, "Energy (eV)", "Counts")
fig.tight_layout(pad=0.8)
fig.savefig(raw_plot_path, bbox_inches="tight")
plt.close(fig)

fig, ax = plt.subplots(figsize=(7.2, 4.8), dpi=300)
fig.patch.set_facecolor("white")
ax.plot(
    df["energy_eV"],
    df["normalized_counts"],
    color="#0d7a64",
    linewidth=2.1,
    solid_capstyle="round",
)
style_axes(ax, "Energy (eV)", "Normalized Intensity (a.u.)", y_max=1.05)
fig.tight_layout(pad=0.8)
fig.savefig(norm_plot_path, bbox_inches="tight")
plt.close(fig)

peak_row = df.loc[df["counts"].idxmax()]
print(f"Saved CSV: {csv_path}")
print(f"Saved raw plot: {raw_plot_path}")
print(f"Saved normalized plot: {norm_plot_path}")
print(
    "Peak intensity: "
    f"{peak_row['counts']:.0f} counts at "
    f"{peak_row['wavelength_nm']:.2f} nm ({peak_row['energy_eV']:.3f} eV)"
)

if __name__ == "__main__":
    main()

```



Python Code for analyzing PL data

```
from __future__ import annotations

import csv
from pathlib import Path

import matplotlib
import numpy as np

matplotlib.use("Agg")
import matplotlib.pyplot as plt

ROOT = Path(__file__).resolve().parent
OUT_DIR = ROOT / "publishable"
OMIT_WINDOW = (500.0, 620.0)

def read_scan(path: Path) -> tuple[dict[str, str], np.ndarray, np.ndarray]:
    metadata: dict[str, str] = {}
    rows: list[tuple[float, float]] = []

    with path.open("r", encoding="utf-8", errors="ignore") as handle:
        for raw_line in handle:
            line = raw_line.strip()
```

```

        if not line:
            continue
        parts = [part.strip() for part in line.split(",")]
        if len(parts) >= 2:
            try:
                rows.append((float(parts[0]), float(parts[1])))
            except ValueError:
                if parts[0] and parts[1]:
                    metadata[parts[0]] = parts[1]

data = np.array(rows, dtype=float)
return metadata, data[:, 0], data[:, 1]

def fit_baseline(x: np.ndarray, y: np.ndarray, exclude: tuple[float, float], degree: int = 3) -> np.ndarray:
    mask = (x < exclude[0]) | (x > exclude[1])
    fit_mask = mask.copy()

    for _ in range(6):
        coeff = np.polyfit(x[fit_mask], y[fit_mask], degree)
        baseline = np.polyval(coeff, x)
        residual = y - baseline
        fit_residual = residual[mask]
        mad = np.median(np.abs(fit_residual - np.median(fit_residual)))
        sigma = 1.4826 * mad if mad > 0 else max(np.std(fit_residual), 1.0)
        fit_mask = mask & (residual < 2.5 * sigma)

    if fit_mask.sum() < degree + 5:
        fit_mask = mask
        break

    coeff = np.polyfit(x[fit_mask], y[fit_mask], degree)
    return np.polyval(coeff, x)

def centered_moving_average(y: np.ndarray, window: int = 5) -> np.ndarray:
    kernel = np.ones(window, dtype=float) / window
    padded = np.pad(y, (window // 2, window // 2), mode="edge")
    return np.convolve(padded, kernel, mode="valid")

def hampel_filter(y: np.ndarray, window: int = 7, n_sigma: float = 4.0) -> np.ndarray:
    out = y.copy()
    half = window // 2

```

```

for i in range(len(y)):
    lo = max(0, i - half)
    hi = min(len(y), i + half + 1)
    segment = y[lo:hi]
    median = np.median(segment)
    mad = np.median(np.abs(segment - median))
    sigma = 1.4826 * mad
    if sigma > 0 and abs(y[i] - median) > n_sigma * sigma:
        out[i] = median

return out

def process_emission_scan(path: Path) -> dict[str, np.ndarray | dict[str, str] | tuple[float, float]]:
    metadata, x, y = read_scan(path)
    exclude = OMIT_WINDOW
    display_ready = y.copy()
    display_mask = (x < exclude[0]) | (x > exclude[1])
    display_ready[display_mask] = hampel_filter(display_ready[display_mask], window=7,
n_sigma=4.0)
    baseline = fit_baseline(x, display_ready, exclude=exclude, degree=3)
    corrected = display_ready - baseline
    corrected = np.where(corrected > 0, corrected, 0.0)
    smooth = centered_moving_average(corrected, window=9)

    mask = (x < exclude[0]) | (x > exclude[1])
    norm = smooth.copy()
    norm_max = np.max(norm[mask]) if np.any(mask) else np.max(norm)
    if norm_max > 0:
        norm = norm / norm_max

    return {
        "metadata": metadata,
        "wavelength_nm": x,
        "raw_counts": y,
        "despiked_counts": display_ready,
        "baseline_counts": baseline,
        "corrected_counts": corrected,
        "smoothed_corrected_counts": smooth,
        "normalized_intensity": norm,
        "scatter_window_nm": exclude,
    }

```

```

def save_cleaned_csv(data: dict[str, dict[str, np.ndarray | tuple[float, float] | dict[str,
str]]]) -> Path:
    OUT_DIR.mkdir(exist_ok=True)
    out = OUT_DIR / "pl_emission_cleaned.csv"
    x = data["scan1"]["wavelength_nm"]

    with out.open("w", newline="", encoding="ascii") as handle:
        writer = csv.writer(handle)
        writer.writerow(
            [
                "wavelength_nm",
                "scan1_raw_counts",
                "scan1_despiked_counts",
                "scan1_baseline_counts",
                "scan1_corrected_counts",
                "scan1_smoothed_corrected_counts",
                "scan1_normalized_intensity",
                "scan2_raw_counts",
                "scan2_despiked_counts",
                "scan2_baseline_counts",
                "scan2_corrected_counts",
                "scan2_smoothed_corrected_counts",
                "scan2_normalized_intensity",
            ]
        )

        for i, wavelength in enumerate(x):
            writer.writerow(
                [
                    f"{wavelength:.1f}",
                    f"{data['scan1']['raw_counts'][i]:.6f}",
                    f"{data['scan1']['despiked_counts'][i]:.6f}",
                    f"{data['scan1']['baseline_counts'][i]:.6f}",
                    f"{data['scan1']['corrected_counts'][i]:.6f}",
                    f"{data['scan1']['smoothed_corrected_counts'][i]:.6f}",
                    f"{data['scan1']['normalized_intensity'][i]:.8f}",
                    f"{data['scan2']['raw_counts'][i]:.6f}",
                    f"{data['scan2']['despiked_counts'][i]:.6f}",
                    f"{data['scan2']['baseline_counts'][i]:.6f}",
                    f"{data['scan2']['corrected_counts'][i]:.6f}",
                    f"{data['scan2']['smoothed_corrected_counts'][i]:.6f}",
                    f"{data['scan2']['normalized_intensity'][i]:.8f}",
                ]
            )

    return out

```

```

def save_notes(data: dict[str, dict[str, np.ndarray | tuple[float, float] | dict[str, str]]]) -> Path:
    out = OUT_DIR / "pl_processing_notes.txt"
    lines = [
        "PL emission processing workflow",
        "1. Parsed instrument-exported text/CSV scans directly",
        "2. Applied a Hampel despiking filter outside the excitation-dominated window",
        "3. Estimated a smooth polynomial baseline outside the excitation-dominated window",
        "4. Saved baseline-subtracted data in the cleaned CSV for reference",
        "5. Final PL figure is diagnostic-style, not UV-Vis-style, because the scan is dominated by excitation/scatter artifacts",
        "6. Top panel shows the raw primary emission scan; bottom panel shows off-band baseline-subtracted intensity only",
        "",
    ]

    for label, key in (("Scan 1", "scan1"), ("Scan 2", "scan2")):
        lo, hi = data[key]["scatter_window_nm"]
        lines.append(f"{label} omitted excitation-dominated window: {lo:.1f}-{hi:.1f} nm")

    out.write_text("\n".join(lines), encoding="ascii")
    return out


def make_plot(data: dict[str, dict[str, np.ndarray | tuple[float, float] | dict[str, str]]]) -> list[Path]:
    plt.rcParams.update(
        {
            "font.size": 11,
            "axes.labelsize": 12,
            "xtick.labelsize": 10,
            "ytick.labelsize": 10,
            "legend.fontsize": 10,
        }
    )

    fig = plt.figure(figsize=(7.6, 6.2))
    gs = fig.add_gridspec(2, 2, height_ratios=[1.0, 0.95], width_ratios=[1.1, 1.0], hspace=0.28, wspace=0.08)
    ax_top = fig.add_subplot(gs[0, :])
    ax_bot_left = fig.add_subplot(gs[1, 0])
    ax_bot_right = fig.add_subplot(gs[1, 1], sharey=ax_bot_left)
    fig.patch.set_facecolor("white")
    ax_top.set_facecolor("white")

```

```

ax_bot_left.set_facecolor("white")
ax_bot_right.set_facecolor("white")

label = "Al-Al$_{2}$-$_{3}$ emission"
x = data["scan1"]["wavelength_nm"]
y_despiked = data["scan1"]["despiked_counts"].copy()
y_raw = data["scan1"]["raw_counts"].copy()
y_clean = data["scan1"]["smoothed_corrected_counts"].copy()
lo, hi = data["scan1"]["scatter_window_nm"]
y_plot = np.where(y_despiked > 0.0, y_despiked, 0.0)
y_clean = np.where(y_clean > 0.0, y_clean, 0.0)

left_mask = x <= lo
right_mask = x >= hi

ax_top.plot(x, y_plot, linewidth=2.1, color="#0b3954", label=label)
ax_top.axvspan(lo, hi, color="#d9d9d9", alpha=0.4, linewidth=0)
ax_top.set_xlim(300, 800)
ax_top.set_ylabel("Raw intensity (counts)")
ax_top.legend(frameon=False, loc="upper right", handlelength=2.4)

ax_bot_left.plot(x[left_mask], y_clean[left_mask], linewidth=2.1, color="#0b3954")
ax_bot_right.plot(x[right_mask], y_clean[right_mask], linewidth=2.1, color="#0b3954")

ymax = max(np.nanmax(y_clean[left_mask]), np.nanmax(y_clean[right_mask]))
ax_bot_left.set_xlim(300, lo)
ax_bot_right.set_xlim(hi, 800)
ax_bot_left.set_ylim(0, ymax * 1.1)

for ax in (ax_top, ax_bot_left, ax_bot_right):
    ax.grid(True, color="#d9d9d9", linewidth=0.8, alpha=0.7)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.tick_params(direction="out", length=4.5, width=1)

ax_bot_right.spines["left"].set_visible(False)
ax_bot_right.tick_params(left=False, labelleft=False)
ax_bot_left.set_ylabel("Off-band PL\n(counts)")
fig.supxlabel("Emission wavelength (nm)", y=0.06)

ax_top.text(
    (lo + hi) / 2,
    ax_top.get_ylim()[1] * 0.93,
    "Excitation-dominated region\n(500-620 nm) excluded",
    ha="center",
    va="top",

```

```

        fontsize=9,
        color="#555555",
    )

    d = 0.015
    kwargs_left = dict(transform=ax_bot_left.transAxes, color="#555555", clip_on=False,
linewidth=1.0)
    kwargs_right = dict(transform=ax_bot_right.transAxes, color="#555555", clip_on=False,
linewidth=1.0)
    ax_bot_left.plot((1 - d, 1 + d), (-d, +d), **kwargs_left)
    ax_bot_left.plot((1 - d, 1 + d), (1 - d, 1 + d), **kwargs_left)
    ax_bot_right.plot((-d, +d), (-d, +d), **kwargs_right)
    ax_bot_right.plot((-d, +d), (1 - d, 1 + d), **kwargs_right)

    ax_top.annotate(
        "Raw scan shown as measured\n(light despiking only)",
        xy=(690, np.interp(690, x, y_plot)),
        xytext=(645, ax_top.get_ylim()[1] * 0.6),
        arrowprops={"arrowstyle": "->", "lw": 1.0, "color": "#555555"},
        fontsize=8.5,
        color="#555555",
    )

    ax_bot_left.text(
        0.02,
        0.93,
        "Off-band PL (baseline-subtracted)",
        transform=ax_bot_left.transAxes,
        fontsize=9,
        color="#555555",
        va="top",
    )

    fig.text(
        0.125,
        0.01,
        "Diagnostic PL figure: top = raw primary scan, bottom = off-band cleaned view;
excitation-dominated region (500-620 nm) excluded",
        ha="left",
        va="bottom",
        fontsize=8.5,
        color="#4d4d4d",
    )
    fig.subplots_adjust(left=0.12, right=0.98, top=0.96, bottom=0.14)

    outputs = []

```

```

for suffix in ("png", "svg", "pdf"):
    target = OUT_DIR / f"pl_emission_publication.{suffix}"
    fig.savefig(target, dpi=300, bbox_inches="tight")
    outputs.append(target)
plt.close(fig)
return outputs

def make_bottom_plot_only(data: dict[str, dict[str, np.ndarray | tuple[float, float] | dict[str, str]]] -> list[Path]:
    plt.rcParams.update(
        {
            "font.size": 11,
            "axes.labelsize": 12,
            "xtick.labelsize": 10,
            "ytick.labelsize": 10,
        }
    )

    fig, (ax_left, ax_right) = plt.subplots(
        1,
        2,
        figsize=(7.6, 3.8),
        sharey=True,
        gridspec_kw={"width_ratios": [1.1, 1.0], "wspace": 0.08},
    )
    fig.patch.set_facecolor("white")
    ax_left.set_facecolor("white")
    ax_right.set_facecolor("white")

    x = data["scan1"]["wavelength_nm"]
    y = np.where(data["scan1"]["smoothed_corrected_counts"] > 0.0,
data["scan1"]["smoothed_corrected_counts"], 0.0)
    lo, hi = data["scan1"]["scatter_window_nm"]

    left_mask = x <= lo
    right_mask = x >= hi

    ax_left.plot(x[left_mask], y[left_mask], linewidth=2.1, color="#0b3954")
    ax_right.plot(x[right_mask], y[right_mask], linewidth=2.1, color="#0b3954")

    ymax = max(np.nanmax(y[left_mask]), np.nanmax(y[right_mask]))
    ax_left.set_xlim(300, lo)
    ax_right.set_xlim(hi, 800)
    ax_left.set_ylim(0, ymax * 1.1)

```

```

for ax in (ax_left, ax_right):
    ax.grid(True, color="#d9d9d9", linewidth=0.8, alpha=0.7)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.tick_params(direction="out", length=4.5, width=1)

ax_right.spines["left"].set_visible(False)
ax_right.tick_params(left=False, labelleft=False)
ax_left.set_ylabel("Off-band PL\n(counts)")
fig.supxlabel("Emission wavelength (nm)", y=0.08)

d = 0.015
kwargs_left = dict(transform=ax_left.transAxes, color="#555555", clip_on=False,
linewidth=1.0)
kwargs_right = dict(transform=ax_right.transAxes, color="#555555", clip_on=False,
linewidth=1.0)
ax_left.plot((1 - d, 1 + d), (-d, +d), **kwargs_left)
ax_left.plot((1 - d, 1 + d), (1 - d, 1 + d), **kwargs_left)
ax_right.plot((-d, +d), (-d, +d), **kwargs_right)
ax_right.plot((-d, +d), (1 - d, 1 + d), **kwargs_right)

ax_left.text(
    0.02,
    0.93,
    "Off-band PL (baseline-subtracted)",
    transform=ax_left.transAxes,
    fontsize=9,
    color="#555555",
    va="top",
)

fig.text(
    0.125,
    0.01,
    "Excitation-dominated region (500-620 nm) excluded",
    ha="left",
    va="bottom",
    fontsize=8.5,
    color="#4d4d4d",
)
fig.subplots_adjust(left=0.12, right=0.98, top=0.95, bottom=0.22)

outputs = []
for suffix in ("png", "svg", "pdf"):
    target = OUT_DIR / f"pl_offband_publication.{suffix}"
    fig.savefig(target, dpi=300, bbox_inches="tight")

```

```

        outputs.append(target)
plt.close(fig)
return outputs

def main() -> None:
    scan1 = process_emission_scan(ROOT / "Emission ScanA1-A1203. Ex 565. BW 5.CSV")
    scan2 = process_emission_scan(ROOT / "Emission ScanA1-A1203. Ex 565. BW 5_second.CSV")
    data = {"scan1": scan1, "scan2": scan2}

    save_cleaned_csv(data)
    save_notes(data)
    outputs = make_plot(data) + make_bottom_plot_only(data)

    for path in outputs:
        print(path)

if __name__ == "__main__":
    main()

```

